

Clinical Findings Questions

Category: Clinical Findings **Total Questions:** 206 **Purpose:** Medical professional review of categorization accuracy

Review Instructions

For each question below, please mark:

- ✓ **Correctly categorized** - Question clearly belongs in this category
- ✗ **Miscategorized** - Question belongs in a different category (note which one)
- ? **Ambiguous** - Question could reasonably fit multiple categories

You don't need to review all questions! Even 20-30 per category is very helpful.

What 'Clinical Findings' Means

Questions asking what findings you would expect to see on physical exam, lab tests, or imaging studies.

Questions for Review

Question 1

Clinical Scenario:

A 60-year-old man presents to the office for a scheduled follow-up visit. He has had hypertension for the past 30 years and his current anti-hypertensive medications include lisinopril (40 mg/day) and hydrochlorothiazide (50 mg/day). He follows most of the lifestyle modifications recommended by his physician, but is concerned about his occasional occipital headaches in the morning. His blood pressure is 160/98 mm Hg. The physician adds another drug to his regimen that acts centrally as an α_2 -adrenergic agonist. Which of the following second messengers is involved in the mechanism of action of this new drug?

Answer Choices:

A. Cyclic adenosine monophosphate B. Cyclic guanosine monophosphate C. Diacylglycerol D. Calcium ions

Correct Answer: A. Cyclic adenosine monophosphate

Question 2

Clinical Scenario:

An 18-year-old man is brought to the emergency department after his mother found him locked in his room stammering about a government conspiracy to brainwash him in subterranean tunnels. His mother says that he has never done this before, but 6 months ago he stopped going to classes and was subsequently suspended from college. She reports that he has become increasingly taciturn over the course of the past month. He drinks one to two beers daily and has smoked one pack of cigarettes daily for 3 years. He occasionally smokes marijuana. His father was diagnosed with schizophrenia at the age of 25 years. The patient has had no friends or social contacts other than his mother since he was suspended. He appears unkempt and aloof. On mental status examination, he is disorganized and shows poverty of speech. He says his mood is "good." He does not hear voices and has no visual or tactile hallucinations. Toxicology screening is negative. Which of the following is a favorable prognostic factor for this patient's condition?

Answer Choices:

A. Predominance of negative symptoms B. Acute onset of symptoms C. Lack of social support D. Cannabis use

Correct Answer: B. Acute onset of symptoms

Question 3

Clinical Scenario:

A 55-year-old postmenopausal woman comes to the physician for a screening Pap smear. She has no history of serious illness. Her last Pap smear was 10 years ago and showed no abnormalities. She has smoked one-half pack of cigarettes daily for 20 years and drinks 3 bottles of wine per week. She is sexually active with multiple male partners and uses condoms inconsistently. Her paternal grandmother had ovarian cancer and her maternal aunt had breast cancer. Pelvic examination shows multiple red, fleshy polypoid masses on the anterior vaginal wall. A biopsy is obtained and histology shows large cells with abundant clear cytoplasm. Which of the following is the most significant risk factor for this diagnosis?

Answer Choices:

A. Alcohol consumption B. Cigarette smoking C. Family history of breast and ovarian cancer D. Diethylstilbestrol exposure in utero

Correct Answer: D. Diethylstilbestrol exposure in utero

Question 4

Clinical Scenario:

A 55-year-old man presents to the family medicine clinic after noticing a gradually enlarging smooth and symmetrical bump on his left forearm at the site of his PPD placement 2 days ago. The patient takes lisinopril for hypertension and metformin for diabetes mellitus type 2. He was screened for tuberculosis 2 days ago as a

requirement for work. He works as a guard at the county prison. He smokes a half-pack of cigarettes per day and has done so for the last 5 years. His heart rate is 88/min, respiratory rate is 16/min, temperature is 37.3°C (99.2°F), and blood pressure is 142/86 mm Hg. The patient appears clean and overweight. The bleb from the screening test is measured at 12 mm. Acid-fast smear of a sputum sample is negative. Which of the following is recommended for the patient at this time?

Answer Choices:

A. Rifampin, isoniazid, pyrazinamide, ethambutol B. Isoniazid for 6 months C. No treatment D. Isoniazid for 9 months

Correct Answer: D. Isoniazid for 9 months

Question 5

Clinical Scenario:

A 35-year-old man comes to the physician because of fatigue and generalized weakness for the past year. He has noticed he has been having fewer bowel movements. He has had pain with defecation and small amounts of blood when wiping. He has not lost weight despite increased efforts to diet and exercise. He has had no fever, throat pain, or difficulty swallowing. His temperature is 36.5°C (97.7°F), pulse is 50/min, blood pressure is 120/90 mm Hg, and BMI is 35 kg/m². Physical examination shows dry skin and a distended abdomen. There is 1+ pitting edema in the lower extremities. On neurological examination, deep tendon reflexes are 1+. Further evaluation of this patient is most likely to show which of the following findings?

Answer Choices:

A. Elevated serum low-density lipoprotein B. Hyperglycemia C. Decreased serum creatinine D. Decreased serum creatine kinase "

Correct Answer: A. Elevated serum low-density lipoprotein

Question 6

Clinical Scenario:

A 74-year-old man is brought to the emergency department after he had copious amounts of blood-stained stools. Minutes later, he turned sweaty, felt light-headed, and collapsed into his wife's arms. Upon admission, he is found to have a blood pressure of 78/40 mm Hg, a pulse of 140/min, and oxygen saturation of 98%. His family history is relevant for both gastric and colorectal cancer. His personal history is relevant for hypertension, for which he takes amlodipine. After an initial successful resuscitation with intravenous fluids, which of the following should be the first step in approaching this case?

Answer Choices:

A. Colonoscopy B. Nasogastric lavage C. Upper endoscopy D. Mesenteric angiography

Correct Answer: B. Nasogastric lavage

Question 7

Clinical Scenario:

A 27-year-old female presents to her OB/GYN for a check-up. During her visit, a pelvic exam and Pap smear are performed. The patient does not have any past medical issues and has had routine gynecologic care with normal pap smears every 3 years since age 21. The results of the Pap smear demonstrate atypical squamous cells of undetermined significance (ASCUS). Which of the following is the next best step in the management of this patient?

Answer Choices:

A. Repeat Pap smear in 1 year B. Repeat Pap smear in 3 years C. Perform an HPV DNA test D. Perform a Loop Electrosurgical Excision Procedure (LEEP)

Correct Answer: C. Perform an HPV DNA test

Question 8

Clinical Scenario:

A 27-year-old homeless man presents to the emergency department with abdominal pain and vomiting. He has a known history of intravenous drug use and has been admitted to the hospital several times before. On physical examination his temperature is 99°F (37.2°C), blood pressure is 130/85 mmHg, pulse is 90/min, respirations are 19/min, and pulse oximetry is 99% on room air. The patient is in obvious discomfort. There is increased salivation and lacrimation. Pupils are reactive to light and 5 mm bilaterally. Cardiopulmonary exam is unremarkable. There is diffuse abdominal tenderness to palpation with no rebound or guarding. Which of the following interventions would have prevented this patient's current condition?

Answer Choices:

A. Buprenorphine B. Naltrexone C. Naloxone D. Bupropion

Correct Answer: A. Buprenorphine

Question 9

Clinical Scenario:

A 53-year-old man comes to the physician because of a 3-month history of a nonpruritic rash. He has been feeling more tired than usual and occasionally experiences pain in his wrists and ankles. He does not smoke or drink alcohol. His temperature is 37.6°C (99.7°F), pulse is 98/min, respirations are 18/min, and blood pressure is 130/75 mm Hg. Physical examination shows multiple, erythematous, purpuric papules on his trunk and extremities that do not blanch when pressed. The remainder of the examination shows no abnormalities. The patient's hemoglobin is 14 g/dL, leukocyte count is 9,500/mm³, and platelet count is 228,000/mm³. Urinalysis and liver function tests are within normal limits. The test for rheumatoid factor is positive. Serum ANA is negative. Serum complement levels are decreased. Serum protein electrophoresis and immunofixation shows

increased gammaglobulins with pronounced polyclonal IgM and IgG bands. Testing for cryoglobulins shows no precipitate after 24 hours. Chest x-ray and ECG show no abnormalities. Which of the following is the most appropriate next step in management?

Answer Choices:

A. Rapid plasma reagin test B. Hepatitis C serology C. pANCA assay D. Bence Jones protein test "

Correct Answer: B. Hepatitis C serology

Question 10

Clinical Scenario:

A 29-year-old woman, gravida 2, para 1, at 10 weeks' gestation comes to the physician for a prenatal visit. Over the past two weeks, she has felt nauseous in the morning and has had vulvar pruritus and dysuria that started 5 days ago. Her first child was delivered by lower segment transverse cesarean section because of macrosomia from gestational diabetes. Her gestational diabetes resolved after the child was born. She appears well. Ultrasound confirms fetal heart tones and an intrauterine pregnancy. Speculum exam shows a whitish chunky discharge. Her vaginal pH is 4.2. A wet mount is performed and microscopic examination is shown. Which of the following is the most appropriate treatment?

Answer Choices:

A. Oral metronidazole B. Intravaginal treatment with lactobacillus C. Oral fluconazole D. Intravaginal clotrimazole

Correct Answer: D. Intravaginal clotrimazole

Question 11

Clinical Scenario:

A 55-year-old female with a history of poorly controlled hyperlipidemia and obesity presents to her primary care physician for a follow-up visit. She reports that she feels well and has no complaints. She currently takes atorvastatin. Her temperature is 99°F (37.2°C), blood pressure is 135/80 mmHg, pulse is 80/min, and respirations are 16/min. Her BMI is 31 kg/m². Her total cholesterol is 290 mg/dl, triglycerides are 120 mg/dl, and LDL cholesterol is 215 mg/dl. Her physician considers starting her on a medication that forces the liver to consume cholesterol to make more bile salts. Which of the following adverse effects is this patient at highest risk of developing following initiation of the medication?

Answer Choices:

A. Gallstones B. Acanthosis nigricans C. Facial flushing D. Fat malabsorption

Correct Answer: D. Fat malabsorption

Question 12

Clinical Scenario:

An investigator is studying the mechanism regulating pigment production in the skin. She has isolated a hormone produced by the anterior and intermediate lobe of the pituitary gland that stimulates neural crest-derived cells to produce pigments through the oxidation and polymerization of the amino acid tyrosine. This hormone is most likely cosecreted with a substance that acts on which of the following receptors?

Answer Choices:

A. Vasopressin receptor B. TSH receptor C. Mu receptor D. Dopamine receptor "

Correct Answer: C. Mu receptor

Question 13

Clinical Scenario:

A 49-year-old woman comes to the physician because of a 1-year history of bloating and constipation alternating with diarrhea. She eats a balanced diet, and there are no associations between her symptoms and specific foods. She had been a competitive swimmer since high school but stopped going to training 4 months ago because her fingers hurt and turned blue as soon as she got into the cold water. She drinks one to two glasses of wine daily. Physical examination shows swollen hands and fingers with wax-like thickening of the skin. There are numerous small, superficial, dilated blood vessels at the tips of the fingers. The abdomen is distended and mildly tender with no guarding or rebound. Further evaluation is most likely to show which of the following findings?

Answer Choices:

A. Bilateral pupillary constriction B. Outpouchings of the sigmoid colon C. Villous atrophy in the duodenum D. Calcium deposits in the skin

Correct Answer: D. Calcium deposits in the skin

Question 14

Clinical Scenario:

A 65-year-old man comes to his primary care physician with a 6-month history of bilateral calf pain. The pain usually occurs after walking his dog a few blocks and is more severe on the right side. He has coronary artery disease, essential hypertension, and type 2 diabetes mellitus. He has smoked two packs of cigarettes daily for 43 years and drinks two alcoholic beverages a day. Current medications include metformin, lisinopril, and aspirin. He is 183 cm (5 ft 11 in) tall and weighs 113 kg (250 lb); BMI is 34.9 kg/m². His temperature is 37°C (98.6°F), pulse is 84/min, and blood pressure is 129/72 mm Hg. Cardiac examination shows a gallop without murmurs. The legs have shiny skin with reduced hair below the knee. Femoral and popliteal pulses are palpable bilaterally. Dorsal pedal pulses are 1+ on the left and absent on the right. Ankle-brachial index (ABI) is

performed in the office. ABI is 0.5 in the right leg, and 0.6 in the left leg. Which of the following is the most appropriate initial step in management?

Answer Choices:

A. Graded exercise therapy B. Propranolol therapy C. Spinal cord stimulation D. Vascular bypass surgery

Correct Answer: A. Graded exercise therapy

Question 15

Clinical Scenario:

A 9-month-old boy is brought to the physician because of increased irritability, continual crying, and fever for 1 day. His mother has noticed that he refuses to lie down on his right side and keeps tugging at his right ear. One week ago, he had a runny nose that has since improved. He was born at term and has been otherwise healthy. He was exclusively breastfed until 2 months of age and is currently bottle-fed with some solid foods introduced. He has been attending a daycare center for the past 5 months. His temperature is 38.4°C (101.1°F) and pulse is 144/min. Otoscopic examination in this child is most likely to show which of the following?

Answer Choices:

A. Bulging erythematous tympanic membrane B. Retracted opacified tympanic membrane C. Vesicles in the ear canal D. Brown mass within the ear canal

Correct Answer: A. Bulging erythematous tympanic membrane

Question 16

Clinical Scenario:

A 69-year-old man is scheduled to undergo radical retropubic prostatectomy for prostate cancer in 2 weeks. He had a myocardial infarction at the age of 54 years. He has a history of GERD, unstable angina, hyperlipidemia, and severe osteoarthritis in the left hip. He is unable to climb up stairs or walk fast because of pain in his left hip. He had smoked one pack of cigarettes daily for 30 years but quit 25 years ago. He drinks one glass of wine daily. Current medications include aspirin, metoprolol, lisinopril, rosuvastatin, omeprazole, and ibuprofen as needed. His temperature is 36.4°C (97.5°F), pulse is 90/min, and blood pressure is 136/88 mm Hg. Physical examination shows no abnormalities. A 12-lead ECG shows Q waves and inverted T waves in leads II, III, and aVF. His B-type natriuretic protein is 84 pg/mL (N < 125). Which of the following is the most appropriate next step in management to assess this patient's perioperative cardiac risk?

Answer Choices:

A. 24-hour ambulatory ECG monitoring B. Radionuclide myocardial perfusion imaging C. No further testing D. Treadmill stress test

Correct Answer: B. Radionuclide myocardial perfusion imaging

Question 17

Clinical Scenario:

A 25-year-old woman presents to her primary care provider for fatigue. She states that she has felt fatigued for the past 6 months and has tried multiple diets and sleep schedules to improve her condition, but none have succeeded. She has no significant past medical history. She is currently taking a multivitamin, folate, B12, iron, fish oil, whey protein, baby aspirin, copper, and krill oil. Her temperature is 98.8°F (37.1°C), blood pressure is 107/58 mmHg, pulse is 90/min, respirations are 13/min, and oxygen saturation is 98% on room air. Laboratory values are as seen below.

Hemoglobin: 8 g/dL Hematocrit: 24% Leukocyte count: 6,500/mm³ with normal differential Platelet count: 147,000/mm³

Physical exam is notable for decreased proprioception in the lower extremities and 4/5 strength in the patient's upper and lower extremities. Which of the following is the best next step in management to confirm the diagnosis?

Answer Choices:

A. Anti-intrinsic factor antibodies B. Bone marrow biopsy C. Iron level D. Transferrin level

Correct Answer: A. Anti-intrinsic factor antibodies

Question 18

Clinical Scenario:

Nine days after being treated for a perforated gastric ulcer and sepsis, a 78-year-old woman develops decreased urinary output and malaise. She required emergency laparotomy and was subsequently treated in the intensive care unit for sepsis. Blood cultures grew *Pseudomonas aeruginosa*. The patient was treated with ceftazidime and gentamicin. She has type 2 diabetes mellitus, arterial hypertension, and osteoarthritis of the hips. Prior to admission, her medications were insulin, ramipril, and ibuprofen. Her temperature is 37.3°C (99.1°F), pulse is 80/min, and blood pressure is 115/75 mm Hg. Examination shows a healing surgical incision in the upper abdomen. Laboratory studies show: Hemoglobin count 14 g/dL Leukocyte count 16,400 mm³ Segmented neutrophils 60% Eosinophils 2% Lymphocytes 30% Monocytes 6% Platelet count 260,000 mm³ Serum Na⁺ 137 mEq/L Cl⁻ 102 mEq/L K⁺ 5.1 mEq/L Urea nitrogen 25 mg/dL Creatinine 4.2 mg/dL Fractional excretion of sodium is 2.1%. Which of the following findings on urinalysis is most likely associated with this patient's condition?"

Answer Choices:

A. Pigmented casts B. Muddy brown casts C. RBC casts D. Waxy casts

Correct Answer: B. Muddy brown casts

Question 19

Clinical Scenario:

A 62-year-old man is referred to neurology by his primary care physician. He is accompanied by his wife. The patient reports having a resting tremor in his left hand for over a year and some "stiffness" in his left arm. His wife notes that he has started to walk "funny" as well. He has a history of hypertension and hyperlipidemia. He takes aspirin, amlodipine, and rosuvastatin. On physical examination, you notice a repetitive circular movement of his left index finger and thumb that resolves with active movement of the hand. Passive motion of the left upper extremity is partially limited by rigidity. Gait is slow and shuffling. The patient is prescribed the most effective treatment for his disorder. Which of the following is the mechanism of a second drug given to prevent adverse effects of this therapy?

Answer Choices:

A. Blocks muscarinic acetylcholine receptors B. Inhibits catechol-O-methyltransferase C. Inhibits aromatic L-amino acid decarboxylase D. Inhibits monoamine oxidase-B

Correct Answer: C. Inhibits aromatic L-amino acid decarboxylase

Question 20

Clinical Scenario:

A 5-month-old male infant from a consanguineous marriage presents with severe sunburns and freckling in sun exposed areas. The mother explains that the infant experiences these sunburns every time the infant goes outside despite applying copious amounts of sunscreen. Which of the following DNA repair mechanisms is defective in this child?

Answer Choices:

A. Nucleotide excision repair B. Base excision repair C. Mismatch repair D. Non-homologous end joining

Correct Answer: A. Nucleotide excision repair

Question 21

Clinical Scenario:

A 29-year-old woman comes to the military physician because of a 2-day history of fever, joint pain, dry cough, chest pain, and a painful red rash on her lower legs. Two weeks ago, she returned from military training in Southern California. She appears ill. Her temperature is 39°C (102.1°F). Physical examination shows diffuse inspiratory crackles over all lung fields and multiple tender erythematous nodules over the anterior aspect of both legs. A biopsy specimen of this patient's lungs is most likely to show which of the following?

Answer Choices:

A. Spherules filled with endospores B. Broad-based budding yeast C. Septate hyphae with acute-angle branching D. Round yeast surrounded by budding yeast cells

Correct Answer: A. Spherules filled with endospores

Question 22

Clinical Scenario:

A 62-year-old man presents to the emergency department with acute pain in the left lower abdomen and profuse rectal bleeding. These symptoms started 3 hours ago. The patient has chronic constipation and bloating, for which he takes lactulose. His family history is negative for gastrointestinal disorders. His temperature is 38.2°C (100.8°F), blood pressure is 90/60 mm Hg, and pulse is 110/min. On physical examination, the patient appears drowsy, and there is tenderness with guarding in the left lower abdominal quadrant. Flexible sigmoidoscopy shows multiple, scattered diverticula with acute mucosal inflammation in the sigmoid colon. Which of the following is the best initial treatment for this patient?

Answer Choices:

A. Dietary modification and antibiotic B. Volume replacement, analgesia, intravenous antibiotics, and endoscopic hemostasis C. Reassurance and no treatment is required D. Elective colectomy

Correct Answer: B. Volume replacement, analgesia, intravenous antibiotics, and endoscopic hemostasis

Question 23

Clinical Scenario:

A 35-year-old woman presents to clinic in emotional distress. She states she has been unhappy for the past couple of months and is having problems with her sleep and appetite. Additionally, she reports significant anxiety regarding thoughts of dirtiness around the house. She states that she cleans all of the doorknobs 5-10 times per day and that, despite her actions, the stress related to cleaning is becoming worse. What is this patient's diagnosis?

Answer Choices:

A. Obsessive compulsive disorder (OCD) B. Tic disorder C. Panic Disorder (PD) D. Generalized anxiety disorder (GAD)

Correct Answer: A. Obsessive compulsive disorder (OCD)

Question 24

Clinical Scenario:

A 60-year-old African-American male with no active medical problems presents to his primary care physician for a general check up. His blood pressure on the previous visit was 145/90, and his blood pressure at this visit is found to be 150/95. He is prescribed hydrochlorothiazide, a thiazide diuretic, to treat his hypertension. The serum level of which of the following is likely to decrease in response to his treatment?

Answer Choices:

A. Cholesterol B. Potassium C. Uric acid D. Calcium

Correct Answer: B. Potassium

Question 25**Clinical Scenario:**

A 74-year-old man comes to the physician for a 6-month history of progressively worsening fatigue and shortness of breath on exertion. He immigrated to the United States 35 years ago from India. His pulse is 89/min and blood pressure is 145/60 mm Hg. Crackles are heard at the lung bases. Cardiac examination shows a grade 3/6 early diastolic murmur loudest at the third left intercostal space. Further evaluation of this patient is most likely to show which of the following?

Answer Choices:

A. Pulsus parvus et tardus B. Fixed splitting of S2 C. Water hammer pulse D. Paradoxical splitting of S2

Correct Answer: C. Water hammer pulse

Question 26**Clinical Scenario:**

A 33-year-old man comes to the otolaryngologist for the evaluation of a 6-month history of difficulty breathing through his nose and clear nasal discharge. He has a history of seasonal atopic rhinosinusitis. Anterior rhinoscopy shows a nasal polyp obstructing the superior nasal meatus. A CT scan of the head is most likely to show opacification of which of the following structures?

Answer Choices:

A. Pterygopalatine fossa and middle ethmoidal sinus B. Sphenoidal sinus and posterior ethmoidal sinuses C. Nasolacrimal duct and eustachian tube D. Frontal sinus and anterior ethmoidal sinus

Correct Answer: B. Sphenoidal sinus and posterior ethmoidal sinuses

Question 27**Clinical Scenario:**

A 62-year-old man comes to the physician because of a growth on his penis that has been gradually increasing in size over the last year. He was diagnosed with HIV 10 years ago. He has been divorced for 25 years and has had "at least 30 sexual partners" since. Physical examination shows a nontender 2.5-cm ulcerated lesion with an erythematous base on the dorsum of the glans. There is firm left inguinal lymphadenopathy. A biopsy of the lesion shows small uniform basophilic cells with central necrosis that invade into the corpus cavernosum. This patient's condition is most likely associated with which of the following pathogens?

Answer Choices:

A. Human papillomavirus B. Epstein-Barr virus C. Haemophilus ducreyi D. Chlamydia trachomatis "

Correct Answer: A. Human papillomavirus

Question 28**Clinical Scenario:**

A 53-year-old man presents to your office with a 2 month history of abdominal bloating. He states that he feels full after eating only a small amount and has experienced bloating, diarrhea, and occasionally vomiting when he tries to eat large amounts. He states his diarrhea has now become more profuse and is altering the quality of his life. One week ago, the patient was given antibiotics for an ear infection. He states he is trying to eat more healthy and has replaced full fat with fat free dairy and is reducing his consumption of meat. His temperature is 99.0°F (37.2°C), blood pressure is 164/99 mmHg, pulse is 85/min, respirations are 14/min, and oxygen saturation is 98% on room air. Laboratory values from a previous office visit are notable for a hemoglobin A1c of 13%. Which of the following is the best treatment of this patient's diarrhea?

Answer Choices:

A. Elimination of dairy from the diet B. Metoclopramide C. Rifaximin D. Vancomycin

Correct Answer: C. Rifaximin

Question 29**Clinical Scenario:**

A 34-year-old man presents to his primary care provider for evaluation of nocturnal cough and dyspnea. He has been a smoker for the past 15 years with a 7.5-pack-year smoking history. He has no significant medical history and takes no medications. His blood pressure is 118/76 mm Hg, the heart rate is 84/min, the respiratory rate is 15/min, and the temperature is 37.0°C (98.6°F). A sputum sample shows Charcot-Leyden crystals and Curschmann spirals. What is the most likely diagnosis?

Answer Choices:

A. Chronic obstructive pulmonary disease B. Atopic asthma C. Panacinar emphysema D. Pneumonia

Correct Answer: B. Atopic asthma

Question 30**Clinical Scenario:**

Which of the following is most likely to have prevented this patient's condition?

Answer Choices:

A. High-fiber diet B. Long-term use of aspirin C. Anticoagulation with warfarin D. Different antibiotic regimen for bronchitis

Correct Answer: A. High-fiber diet

Question 31

Clinical Scenario:

A 70-year-old man presents with right-sided weakness. The patient says that symptoms acutely onset an hour ago while watching television at home. Past medical history is significant for poorly controlled hypertension and 2 myocardial infarctions in the last 2 years. His blood pressure is 140/100 mm Hg, the respiratory rate is 18/min, and the heart rate is 58/min. On physical examination, strength is 1/5 in the right upper and lower extremities. A noncontrast CT of the head is shown in the image. The physician explains that this patient's condition is most likely caused by his poorly controlled hypertension. Which of the following conditions can also cause a similar kind of lesion?

Answer Choices:

A. Cerebral atrophy B. Saccular aneurysm C. Amyloid angiopathy D. Pterion fracture

Correct Answer: C. Amyloid angiopathy

Question 32

Clinical Scenario:

A 47-year-old woman comes to the physician because of involuntary leakage of urine for the past 4 months, which she has experienced when bicycling to work and when laughing. She has not had any dysuria or urinary urgency. She has 4 children that were all delivered vaginally. She is otherwise healthy and takes no medications. The muscles most likely affected by this patient's condition receive efferent innervation from which of the following structures?

Answer Choices:

A. S1-S2 nerve roots B. Superior hypogastric plexus C. Obturator nerve D. S3-S4 nerve roots

Correct Answer: D. S3-S4 nerve roots

Question 33

Clinical Scenario:

An 80-year-old man is admitted to the hospital after the sudden onset of sub-sternal chest pain and shortness of breath while sitting in a chair. He has hypertension and type 2 diabetes mellitus. He has smoked 1 pack of cigarettes daily for 42 years. Four days after admission, he becomes tachycardic and then loses consciousness; the cardiac monitor shows irregular electrical activity. Cardiac examination shows a new systolic murmur at the

apex. Despite appropriate measures, he dies. Microscopic evaluation of the myocardium is most likely to show which of the following?

Answer Choices:

A. Coagulative necrosis with dense neutrophilic infiltrate B. Wavy myocardial fibers without inflammatory cells C. Dense granulation tissue with collagenous scar formation D. Hyperemic granulation tissue with abundance of macrophages

Correct Answer: D. Hyperemic granulation tissue with abundance of macrophages

Question 34

Clinical Scenario:

A 14-year-old boy is brought to the emergency department from school after falling in gym class. He was unable to stand after the accident and has a painful and swollen knee. On presentation he says that he has never had an incident like this before; however, he does suffer from hard to control nosebleeds and prolonged bleeding after minor cuts. Based on his presentation a panel of bleeding tests is obtained with the following results:

Bleeding time: Prolonged Prothrombin time: Normal Partial thromboplastin time: Prolonged

Which of the following describes the function of the component that is defective in the most likely cause of this patient's symptoms?

Answer Choices:

A. Binds to a nucleotide derivative B. Binds to subendothelial collagen C. Catalyzes the conversion of factor X D. It is a cofactor for an epoxide reductase

Correct Answer: B. Binds to subendothelial collagen

Question 35

Clinical Scenario:

A 40-year-old man presents to the emergency department with a chief complaint of chest pain for the last 3 hours. His ECG shows normal sinus rhythm with ST-segment elevation in leads II, III, and aVF and reciprocal segment depression in leads V1–V6. On physical examination, cardiac sounds are normal on auscultation. His blood pressure is 92/64 mm Hg and heart rate was 93/min. A tissue plasminogen activator is administered to the patient intravenously within 1 hour of hospital arrival due to a lack of available percutaneous coronary intervention. After 6 hours of therapy, the patient's clinical condition starts to deteriorate. ECG on the monitor shows accelerated idioventricular rhythm, which within a couple of minutes changes to ventricular fibrillation. Before any measures could be started, the patient deteriorates further and must be transferred to the ICU. What is the most likely etiology of the ECG findings in this patient?

Answer Choices:

A. Increase in cellular pH B. Calcium efflux C. Inhibition of lipid peroxidation D. Free radical formation

Correct Answer: D. Free radical formation

Question 36

Clinical Scenario:

A 52-year-old man is brought to the emergency department by a friend because of a 5-day history of fever and cough productive of purulent sputum. One week ago, he was woken up by an episode of heavy coughing while lying on his back. He drinks large amounts of alcohol daily and has spent most of his time in bed since his wife passed away 2 months ago. His temperature is 38°C (100.4°F), pulse is 96/min, respirations are 24/min, and blood pressure is 110/84 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 87%. Physical examination shows poor dentition and swollen gums. A CT scan of the chest is most likely to show a pulmonary infiltrate in which of the following locations?

Answer Choices:

A. Posterior basal segment of the right lower lobe B. Apicoposterior segment of the left upper lobe C. Superior segment of the right lower lobe D. Posterior basal segment of the left lower lobe

Correct Answer: C. Superior segment of the right lower lobe

Question 37

Clinical Scenario:

A 9-year-old boy is referred to an orthopedic surgeon after his primary care physician noticed that he was developing scoliosis. He has been otherwise healthy. His family history includes blindness and a cancer causing extremely high blood pressure. On physical exam there are scattered nodules in his skin as well as the findings shown in the photographs. This patient's disorder most likely exhibits which of the following modes of inheritance?

Answer Choices:

A. Autosomal dominant B. Autosomal recessive C. X-linked dominant D. X-linked recessive

Correct Answer: A. Autosomal dominant

Question 38

Clinical Scenario:

A 2-year-old female with abdominal pain undergoes laparoscopic surgery. An outpouching of tissue is excised from the ileum and sent to the laboratory for evaluation. The pathologist notes inflammation and the presence of mucosa, submucosa, and muscle in the walls of the specimen. Which of the following is the most likely diagnosis?

Answer Choices:

A. Crohn's disease B. Meckel's diverticulum C. Appendicitis D. Henoch-Schonlein purpura

Correct Answer: B. Meckel's diverticulum

Question 39**Clinical Scenario:**

A 5-year-old male is brought to the pediatrician by his mother, who relates a primary complaint of a recent history of five independent episodes of vomiting over the last 10 months, most recently 3 weeks ago. Each time, he has awoken early in the morning appearing pale, feverish, lethargic, and complaining of severe nausea. This is followed by 8-12 episodes of non-bilious vomiting over the next 24 hours. Between these episodes he returns to normal activity. He has no significant past medical history and takes no other medications. Review of systems is negative for changes in vision, gait disturbance, or blood in his stool. His family history is significant only for migraine headaches. Vital signs and physical examination are within normal limits. Initial complete blood count, comprehensive metabolic panel, and abdominal radiograph were unremarkable. What is the most likely diagnosis?

Answer Choices:

A. Intracranial mass B. Cyclic vomiting syndrome C. Gastroesophageal reflux D. Intussusception

Correct Answer: B. Cyclic vomiting syndrome

Question 40**Clinical Scenario:**

A 45-year-old woman comes to the physician because of progressive difficulty swallowing solids and liquids over the past 4 months. She has lost 4 kg (9 lb) during this period. There is no history of serious illness. She emigrated to the US from Panama 7 years ago. She does not smoke cigarettes or drink alcohol. Cardiopulmonary examination shows a systolic murmur and an S3 gallop. A barium radiograph of the chest is shown. Endoscopic biopsy of the distal esophagus is most likely to show which of the following?

Answer Choices:

A. Atrophy of esophageal smooth muscle cells B. Infiltration of eosinophils in the epithelium C. Absence of myenteric plexus neurons D. Presence of metaplastic columnar epithelium

Correct Answer: C. Absence of myenteric plexus neurons

Question 41**Clinical Scenario:**

A 33-year-old woman comes to the physician because of a 4-day history of fever, anterior neck pain, and throat swelling. She has no history of serious illness. Her temperature is 38.1°C (100.6°F) and pulse is 109/min. Physical examination shows diaphoresis and a fine tremor of the outstretched hands. The thyroid gland is enlarged, firm, and tender to palpation. Serum thyroid stimulating hormone level is 0.06 µU/mL and erythrocyte sedimentation rate is 65 mm/h. 123I scan shows an enlarged thyroid gland with diffusely decreased uptake. Histologic examination of a thyroid biopsy specimen is most likely to show which of the following findings?

Answer Choices:

A. Follicular epithelial cell hyperplasia B. Concentric intracellular lamellar calcifications C. Lymphocytic infiltration with germinal follicle formation D. Noncaseating granulomas with multinucleated giant cells

Correct Answer: D. Noncaseating granulomas with multinucleated giant cells

Question 42

Clinical Scenario:

A 68-year-old male with past history of hypertension, hyperlipidemia, and a 30 pack/year smoking history presents to his primary care physician for his annual physical. Because of his age and past smoking history, he is sent for screening abdominal ultrasound. He is found to have a 4 cm infrarenal abdominal aortic aneurysm. Surgical repair of his aneurysm is indicated if which of the following are present?

Answer Choices:

A. Abdominal, back, or groin pain B. Smoking history C. Growth of < 0.5 cm in one year D. Marfan's syndrome

Correct Answer: A. Abdominal, back, or groin pain

Question 43

Clinical Scenario:

A 24-year-old woman comes to her primary care physician because she has not had a menstrual period for 6 months. She is a competitive runner and has been training heavily for the past year in preparation for upcoming races. She has no family or personal history of serious illness. She has not been sexually active for the past 9 months. Her temperature is 36.9°C (98.4° F), pulse is 51/min, respirations are 12/min, and blood pressure is 106/67 mm Hg. Her BMI is 18.1 kg/m². Which of the following is the most likely cause of her amenorrhea?

Answer Choices:

A. Increased prolactin secretion B. Decreased frequency of GnRH release from the hypothalamus C. Intrauterine adhesions D. Increased LH release and increased ovarian androgen production

Correct Answer: B. Decreased frequency of GnRH release from the hypothalamus

Question 44

Clinical Scenario:

A 14-year-old girl is brought by her parents to the physician because of recurrent episodes of shortness of breath and nonproductive cough over the past 3 months. She has had two episodes per week, which have resolved spontaneously with rest. Once a month, she wakes up at night with shortness of breath. Spirometry shows an FVC of 95% and an FEV1 of 85% of predicted. Treatment with inhaled budesonide-formoterol as needed is begun. Two weeks later, she is brought to the physician with acute onset of dyspnea and wheezing. Her pulse is 95/min and respirations are 32/min. Which of the following is the most appropriate initial pharmacotherapy?

Answer Choices:

A. Albuterol B. Guaifenesin C. Montelukast sodium D. Fluticasone

Correct Answer: A. Albuterol

Question 45

Clinical Scenario:

A 44-year-old woman comes to the physician because of progressively worsening shortness of breath with exertion and intermittent palpitations over the last 2 months. She has had neither chest pain nor a cough. Her pulse is 124/min and irregular. Physical examination shows a grade 4/6 high-pitched holosystolic murmur that is best heard at the apex and radiates to the back. The murmur increases in intensity when she clenches her hands into fists. The lungs are clear to auscultation. Further evaluation of this patient is most likely to show which of the following findings?

Answer Choices:

A. Pulmonary artery thrombus on computed tomography scan B. Obstruction of the right marginal artery on coronary angiogram C. Reversible area of myocardial ischemia on nuclear stress test D. Dilation of left atrium on echocardiogram

Correct Answer: D. Dilation of left atrium on echocardiogram

Question 46

Clinical Scenario:

A 25-day-old newborn is brought to the pediatrician for lethargy, poor muscle tone, and feeding difficulty with occasional regurgitation that recently turned into projectile vomiting. The child was born via vaginal delivery without complications. On examination, the vital signs include: pulse 130/min, respiratory rate 30/min, blood pressure 96/60 mm Hg, and temperature 36.5°C (97.7°F). The physical examination shows a broad nasal bridge, oral thrush, hepatosplenomegaly, and generalized hypotonia. Further tests of blood and urine samples help the pediatrician to diagnose the child with an enzyme deficiency. More extensive laboratory testing reveals

normal levels of citrulline and hypoglycemia. There are also elevated levels of ketone bodies, glycine, and methylmalonic acid. Which of the following is the product of the reaction catalyzed by the deficient enzyme in this patient?

Answer Choices:

A. Pyruvate B. Succinyl-CoA C. Methylmalonyl-CoA D. Acetyl-CoA

Correct Answer: B. Succinyl-CoA

Question 47

Clinical Scenario:

A 24-year-old woman arrives to an urgent care clinic for "eye discoloration." She states that for the past 3 days she has had the "stomach flu" and has not been eating much. Today, she reports she is feeling better, but when she woke up "the whites of [her] eyes were yellow." She denies fever, headache, palpitations, abdominal pain, nausea, vomiting, and diarrhea. She was recently diagnosed with polycystic ovary syndrome during a gynecology appointment 2 weeks ago for irregular menses. Since then, she has been taking a daily combined oral contraceptive. She takes no other medications. Her temperature is 98.6°F (37°C), blood pressure is 120/80 mmHg, and pulse is 76/min. Body mass index is 32 kg/m². On physical examination, there is scleral icterus and mild jaundice. Liver function tests are drawn, as shown below:

Alanine aminotransferase (ALT): 19 U/L Aspartate aminotransferase (AST): 15 U/L Alkaline phosphatase: 85 U/L Albumin: 4.0 g/dL Total bilirubin: 12 mg/dL Direct bilirubin: 10 mg/dL Prothrombin time: 13 seconds

If a liver biopsy were to be performed and it showed a normal pathology, which of the following would be the most likely diagnosis?

Answer Choices:

A. Cholelithiasis B. Crigler-Najjar syndrome C. Dubin-Johnson syndrome D. Rotor syndrome

Correct Answer: D. Rotor syndrome

Question 48

Clinical Scenario:

A 28-year-old woman comes to the emergency department because of a 2-day history of dark urine, increasing abdominal pain, and a tingling sensation in her arms and legs. She has a history of epilepsy. Her current medication is phenytoin. She is nauseated and confused. Following the administration of hemin and glucose, her symptoms improve. The beneficial effect of this treatment is most likely due to inhibition of which of the following enzymes?

Answer Choices:

A. Aminolevulinic acid synthase B. Ferrochelatase C. Porphobilinogen deaminase D. Uroporphyrinogen decarboxylase

Correct Answer: A. Aminolevulinate acid synthase

Question 49

Clinical Scenario:

A mother presents to the family physician with her 16-year-old son. She explains, "There's something wrong with him doc. His grades are getting worse, he's cutting class, he's gaining weight, and his eyes are often bloodshot." Upon interviewing the patient apart from his mother, he seems withdrawn and angry at times when probed about his social history. The patient denies abuse and sexual history. What initial test should be sent to rule out the most likely culprit of this patient's behavior?

Answer Choices:

A. Complete blood count B. Blood culture C. Sexually transmitted infection (STI) testing D. Urine toxicology screen

Correct Answer: D. Urine toxicology screen

Question 50

Clinical Scenario:

A 38-year-old woman comes to the physician because of a 10-month history of nonbloody diarrhea and recurrent episodes of flushing and wheezing. She does not take any medications. Physical examination shows a hyperpigmented rash around the base of her neck. Cardiac examination shows a grade 4/6, holosystolic murmur in the 5th intercostal space at the left midclavicular line. Echocardiography shows left-sided endocardial and valvular fibrosis with moderate mitral regurgitation; there are no septal defects or right-sided valvular defects. Urinalysis shows increased 5-hydroxyindoleacetic acid concentration. Further evaluation of this patient is most likely to show which of the following findings?

Answer Choices:

A. Tumor in the pancreas without metastasis B. Tumor in the lung without metastasis C. Tumor in the appendix without metastasis D. Tumor in the descending colon with hepatic metastasis

Correct Answer: B. Tumor in the lung without metastasis

Question 51

Clinical Scenario:

A 32-year-old woman comes to the physician for genetic consultation. She has a history of recurrent generalized seizures, diffuse muscular weakness, and multiple episodes of transient left-sided paresis. She has been hospitalized several times for severe lactic acidosis requiring intravenous fluid hydration. Her 10-year-old daughter also has seizures and muscle weakness. Her 7-year-old son has occasional muscle weakness and

headaches but has never had a seizure. Pathologic examination of a biopsy specimen from the woman's soleus muscle shows ragged-appearing muscle fibers. Genetic analysis of the patient's son is most likely to show which of the following?

Answer Choices:

A. Silenced paternal gene copy B. Heterogenous mitochondrial DNA C. Mutation in DNA repair gene D. Genetically distinct cell lines

Correct Answer: B. Heterogenous mitochondrial DNA

Question 52

Clinical Scenario:

A 56-year-old woman comes to the physician because of increasing muscle weakness in her shoulders and legs for 1 month. She has difficulties standing up and combing her hair. She also has had a skin rash on her face and hands for the past week. She has hypercholesterolemia treated with simvastatin. She has chronic eczema of her feet that is well-controlled with skin moisturizer and corticosteroid cream. Her mother and sister have thyroid disease. Vital signs are within normal limits. Examination shows facial erythema. A photograph of her hands is shown. Muscle strength is 3/5 in the iliopsoas, hamstring, deltoid, and biceps muscles. Sensation to pinprick, temperature, and vibration is intact. Further evaluation of this patient is most likely to show which of the following?

Answer Choices:

A. Symptom resolution on statin withdrawal B. Pathological edrophonium test C. Intramuscular inclusion bodies D. Elevated serum CA-125

Correct Answer: D. Elevated serum CA-125

Question 53

Clinical Scenario:

A microbiology student was given a swab containing an unknown bacteria taken from the wound of a soldier and asked to identify the causative agent. She determined that the bacteria was a gram-positive, spore-forming bacilli, but had difficulty narrowing it down to the specific bacteria. The next test she performed was the Nagler's test, in which she grew the bacteria on a plate made from egg yolk, which would demonstrate the ability of the bacteria to hydrolyze phospholipids and produce an area of opacity. Half the plate contained a specific antitoxin which prevented hydrolysis of phospholipids while the other half did not contain any antitoxin. The bacteria produced an area of opacity only on half of the plate containing no antitoxin. Which of the following toxins was the antitoxin targeting?

Answer Choices:

A. Alpha toxin B. Tetanus toxin C. Exotoxin A D. Diphtheria toxin

Correct Answer: A. Alpha toxin

Question 54

Clinical Scenario:

A 17-year-old girl makes an appointment with her pediatrician because she is concerned that she may have gotten a sexually transmitted infection. Specifically, she had unprotected sex two weeks ago and has since been experiencing painful urination and abdominal pain. Laboratory tests confirm a diagnosis of Chlamydial infection. At this point, the girl says that she wants to personally give permission to be treated rather than seek consent from her parents because they do not know that she is in a relationship. She also asks that the diagnosis not be reported to anyone. What should the physician do with regards to these two patient requests?

Answer Choices:

A. Choose based on the physician's interpretation of the patient's best interests B. Contact her parents as well as report to public health agencies C. Do not contact her parents and do not report to public health agencies D. Do not contact her parents but do report to public health agencies

Correct Answer: D. Do not contact her parents but do report to public health agencies

Question 55

Clinical Scenario:

A study is designed to assess the functions of immune components. The investigator obtains a lymph node biopsy from a healthy subject and observes it under a microscope. A photomicrograph of the cross-section of this lymph node is shown. Which of the following immunologic processes most likely occurs in the region labeled with an arrow?

Answer Choices:

A. Macrophage activation B. Negative selection C. V(D)J recombination D. Isotype switching

Correct Answer: D. Isotype switching

Question 56

Clinical Scenario:

A 26-year-old woman seeks evaluation at an urgent care clinic with complaints of fever and generalized muscle and joint pain for the past 3 days. She also complains of nausea, but denies vomiting. She does not mention any past similar episodes. Her past medical history is unremarkable, but she returned to the United States 1 week ago after spending 2 weeks in southeast Asia doing charity work. She received all the recommended vaccines prior to traveling. The temperature is 40.0°C (104.0°F), the respirations are 15/min, the pulse is 107/min, and the blood pressure is 98/78 mm Hg. Physical examination shows mild gingival bleeding and a petechial rash over the trunk. Laboratory studies show the following: Laboratory test Leukocyte count 4,000/mm³ Platelet

count 100,000/mm³ Partial thromboplastin time (activated) 45 seconds Which of the following is the most likely cause of this patient's condition?

Answer Choices:

A. Dengue fever B. Ebola virus C. Leptospirosis D. Yellow fever

Correct Answer: A. Dengue fever

Question 57

Clinical Scenario:

A team of intensivists working in a private intensive care unit (ICU) observe that the clinical efficacy of vancomycin is low, and proven nosocomial infections have increased progressively over the past year. A clinical microbiologist is invited to conduct a bacteriological audit of the ICU. He analyzes the microbiological reports of all patients treated with vancomycin over the last 2 years and takes relevant samples from the ICU for culture and antibiotic sensitivity analysis. The audit concludes that there is an increased incidence of vancomycin-resistant *Enterococcus fecalis* infections. Which of the following mechanisms best explains the changes that took place in the bacteria?

Answer Choices:

A. Protection of the antibiotic-binding site by Qnr protein B. Replacement of the terminal D-ala in the cell wall peptidoglycan by D-lactate C. Increased expression of efflux pumps which extrude the antibiotic from the bacterial cell D. Decreased number of porins in the bacterial cell wall leading to decreased intracellular entry of the antibiotic

Correct Answer: B. Replacement of the terminal D-ala in the cell wall peptidoglycan by D-lactate

Question 58

Clinical Scenario:

A 27-year-old woman presents to the emergency department because of muscle tightness and pain. She says that she has experienced increasing tightness and cramping of the muscles on the left side of her neck. She also says that she has trouble looking downwards because her "eyes are stuck." She has a history of schizophrenia, which is being treated with haloperidol. Her temperature is 37.0°C (98.6°F), the pulse is 110/min, the respirations are 18/min, and the blood pressure is 115/71 mm Hg. Physical examination shows significant stiffness in her neck with muscle spasms. Her head is tilted severely to the left side, and her eyes are steady in upward gaze. Respiratory examination shows good air entry bilaterally with no wheezing. Which of the following medicines is the most appropriate next step in management?

Answer Choices:

A. Benztropine B. Dantrolene C. Haloperidol D. Propranolol

Correct Answer: A. Benztropine

Question 59

Clinical Scenario:

A 22-year-old primigravid woman comes to the physician for her initial prenatal visit at 12 weeks' gestation. She has had generalized fatigue and shortness of breath over the past 2 months. She has also had a tingling sensation in her toes for the past month. Three years ago, she was treated for gonorrhea. She follows a strict vegan diet since the age of 13 years. Her temperature is 37°C (98.6°F), pulse is 111/min, and blood pressure is 122/80 mm Hg. Examination shows pale conjunctivae and a shiny tongue. Muscle tone and strength is normal. Deep tendon reflexes are 2+ bilaterally. Sensation to vibration and position is decreased over the upper and lower extremities. When asked to stand, hold her arms in front of her, and close her eyes, she loses her balance and takes a step backward. Which of the following is most likely to have prevented this patient's condition?

Answer Choices:

A. Calcium supplementation B. Vitamin B12 supplementation C. Thyroxine supplementation D. Penicillin G therapy

Correct Answer: B. Vitamin B12 supplementation

Question 60

Clinical Scenario:

An investigator studying the immunologic profile of various cells notices that the blood of a test subject agglutinates upon addition of a serum containing antibodies against P blood group antigens. Infection with which of the following pathogens would most likely be prevented by these antibodies?

Answer Choices:

A. Parvovirus B19 B. Babesia microti C. Plasmodium vivax D. Influenza virus

Correct Answer: A. Parvovirus B19

Question 61

Clinical Scenario:

A 47-year-old patient returns to his primary care physician after starting aspirin two weeks ago for primary prevention of coronary artery disease. He complains that he wakes up short of breath in the middle of the night and has had coughing "attacks" three times. After discontinuing aspirin, what medication is most appropriate for prevention of similar symptoms in this patient?

Answer Choices:

A. Tiotropium B. Prednisone C. Montelukast D. Fluticasone

Correct Answer: C. Montelukast

Question 62

Clinical Scenario:

A 2800-g (6-lb 3-oz) male newborn is born at 39 weeks' gestation to a 22-year-old woman, gravida 2, para 2, after an uncomplicated labor and delivery. The mother did not receive prenatal care. She traveled to Brazil to visit relatives during the first trimester of her pregnancy. She has bipolar disorder treated with lithium. The newborn is at the 50th percentile for height, 25th percentile for weight, and 2nd percentile for head circumference. Neurologic examination shows spasticity of the upper and lower extremities. The wrists are fixed in flexion bilaterally. Deep tendon reflexes are 4+ and symmetric. Ophthalmoscopic examination shows focal pigmentary retinal mottling. Testing for otoacoustic emissions is negative. Which of the following measures during the mother's pregnancy is most likely to have prevented this newborn's condition?

Answer Choices:

- A. Avoid consumption of undercooked meat
- B. Use of mosquito repellent
- C. Daily intake of prenatal vitamins
- D. Discontinuation of mood stabilizer

Correct Answer: B. Use of mosquito repellent

Question 63

Clinical Scenario:

A 48-year-old woman presents to the emergency room because of severe back pain after a fall. She says that she was walking home from work when she slipped on a patch of ice on the sidewalk. Since she did not have anything to hold onto, she fell backwards and landed on her posterior iliac crests bilaterally. Immediately after the fall, she started experiencing back pain and tenderness that concerned her enough to call for an ambulance. Her past medical history is significant for arthritis, diabetes, and hypertension. On arrival, her temperature is 99°F (37.2°C), blood pressure is 129/86 mmHg, pulse is 112/min, respirations are 19/min. Physical exam reveals tenderness to palpation over the middle of her lower back. A drug that may have predisposed this patient to this outcome most likely has which of the following mechanisms?

Answer Choices:

- A. Inhibition of leukotriene and prostaglandin production
- B. Inhibition of circulating cytokine
- C. Inhibition of folate processing
- D. Inhibition of prostaglandin production alone

Correct Answer: A. Inhibition of leukotriene and prostaglandin production

Question 64

Clinical Scenario:

A 52-year-old man awakens in the middle of the night with excruciating pain in his right great toe. He reports that even the touch of the bed sheet was unbearably painful. His right foot is shown in figure A. He is treated with colchicine. Which of the following describes the mechanism of colchicine?

Answer Choices:

A. Inhibition of xanthine oxidase B. Inhibition of reabsorption of uric acid in proximal convoluted tubule C. Binds to glucocorticoid receptor D. Decreases microtubule polymerization

Correct Answer: D. Decreases microtubule polymerization

Question 65

Clinical Scenario:

An investigator is studying the rate of wound healing by secondary intention. He performs a biopsy of a surgically debrided wound 1 day and 5 days after the initial surgical procedure. The second biopsy shows wound contraction, endothelial cell proliferation, and accumulation of macrophages. The cells responsible for wound contraction also secrete a protein that assembles in supercoiled triple helices. The protein type secreted by these cells is most abundant in which of the following structures?

Answer Choices:

A. Reticular fibers B. Nucleus pulposus C. Basal lamina D. Corneal stroma

Correct Answer: A. Reticular fibers

Question 66

Clinical Scenario:

A 19-year-old primigravid woman at 32 weeks' gestation comes to the physician because of a 2-day history of headache and blurred vision. She has had no prenatal care. She is diagnosed with pre-eclampsia. Amniocentesis shows a lecithin-sphingomyelin ratio of 0.7. If delivery is induced at this time, the newborn is most likely to show which of the following findings?

Answer Choices:

A. Decreased right ventricular afterload B. Increased diffusion capacity for carbon monoxide C. Decreased functional residual capacity D. Increased lung compliance

Correct Answer: C. Decreased functional residual capacity

Question 67

Clinical Scenario:

A 15-year-old boy is brought to the emergency department by his father 10 minutes after falling into a frozen lake during ice fishing. He was in the water for less than 1 minute before his father managed to pull him out. On arrival, his clothes are still wet and he appears scared. His body temperature is 36.2°C (97.1°F), pulse is 102/min, blood pressure is 133/88 mm Hg. Which of the following mechanisms contributes most to maintaining this patient's core body temperature?

Answer Choices:

A. Involuntary muscular contractions B. Contraction of arrector pili muscles C. Inhibition of the thyroid axis
D. Activation of thermogenin

Correct Answer: A. Involuntary muscular contractions

Question 68**Clinical Scenario:**

A 54-year-old woman comes to the physician because of constant dull pain, swelling, and progressive stiffness of the right knee for 3 days. Use of over-the-counter analgesics has only provided minimal relief of her symptoms. She has not had any similar symptoms in the past. She takes hydrochlorothiazide for hypertension. Examination of the right knee shows a large effusion and mild erythema. There is moderate tenderness to palpation. Range of motion is limited by pain. Arthrocentesis of the right knee is performed, and microscopic examination of the synovial fluid under polarized light is shown. Further evaluation of this patient is most likely to show which of the following findings?

Answer Choices:

A. Human leukocyte antigen-B27 positivity B. Knee joint space narrowing with subchondral sclerosis C. Calcification of the meniscal cartilage D. Chalky nodules on the external ear

Correct Answer: C. Calcification of the meniscal cartilage

Question 69**Clinical Scenario:**

A 44-year-old man comes to the physician because of a 2-week history of lower extremity swelling and frothy urine. He has a history of chronic hepatitis C infection. Physical examination shows 3+ pitting edema of the lower legs and ankles. Further evaluation of this patient is most likely to show which of the following?

Answer Choices:

A. Decreased blood urea nitrogen B. Increased lipoproteins C. Decreased cystatin C D. Increased antithrombin III

Correct Answer: B. Increased lipoproteins

Question 70**Clinical Scenario:**

A 43-year-old man comes to the physician for a 1-week history of swelling around his eyes and decreased urination. His pulse is 87/min, and blood pressure is 152/95 mm Hg. Physical examination shows 1+ periorbital and pretibial edema. Serum studies show a urea nitrogen concentration of 21 mg/dL and a creatinine

concentration of 1.4 mg/dL. Urinalysis shows 3+ blood and 1+ protein. Further evaluation of this patient is most likely to show which of the following?

Answer Choices:

A. Urinary rhomboid crystals B. Hypoalbuminemia C. Red blood cell casts D. Detached renal tubular epithelial cells

Correct Answer: C. Red blood cell casts

Question 71

Clinical Scenario:

A 2-year-old boy is brought to the physician because of progressive difficulty breathing and a productive cough for the past 2 days. During the past 6 months, he has had recurrent episodes of pneumonia treated with antibiotics. He has not gained weight in this time period. His temperature is 38.5°C (101.3°F), pulse is 130/min, respirations are 18/min, and blood pressure is 100/60 mm Hg. Auscultation of the lungs shows decreased breath sounds over the right lung fields. Ocular examination shows periorbital subcutaneous bleeding and bulging of both eyes. His leukocyte count is 16,000/mm³. An x-ray of the chest shows a right-sided opacity and a collapsed right lung. An MRI of the chest shows a heterogeneous mass in the posterior mediastinum that compresses the heart and the great vessels to the left side. Further evaluation is most likely to show which of the following?

Answer Choices:

A. Overexpression of the N-myc oncogene B. Increased lymphoblast count in the bone marrow C. Unregulated B-cell proliferation in the mediastinum D. Acid-fast bacteria on sputum microscopy

Correct Answer: A. Overexpression of the N-myc oncogene

Question 72

Clinical Scenario:

A 45-year-old woman comes to the physician because of a 5-kg (11-lb) weight loss and difficulty swallowing. She is able to swallow liquids without difficulty but feels like solid foods get stuck in her throat. Physical examination shows taut skin and limited range of motion of the fingers. There are telangiectasias over the cheeks. An esophageal motility study shows absence of peristalsis in the lower two-thirds of the esophagus and decreased lower esophageal sphincter pressure. Further evaluation of this patient is most likely to show which of the following?

Answer Choices:

A. Amyloid deposits in the liver B. Budding yeasts on the oral mucosa C. Parasite nests in the myocardium D. Arteriolar wall thickening in the kidney

Correct Answer: D. Arteriolar wall thickening in the kidney

Question 73

Clinical Scenario:

A 6-year-old boy is brought to the pediatrician by his mother complaining of abdominal pain and constipation. She reports that his appetite has been reduced and that he has not had a bowel movement in 2 days. Prior to this, he had a regular bowel movement once a day. She also reports that he has appeared to be more tired than usual. The family recently moved into a house built in the 1940s and have just begun renovations. The child was born via spontaneous vaginal delivery at 39 weeks gestation. He is up to date on all vaccinations and meeting all developmental milestones. Today, his blood pressure is 120/80 mm Hg, heart rate is 95/min, respiratory rate is 25/min, and temperature is 37.0°C (98.6°F). A physical exam is only significant for moderate conjunctival pallor. A peripheral blood smear shows red blood cells with basophilic stippling. What is the most likely mechanism causing this patient's symptoms?

Answer Choices:

A. Inhibition of ferrochelatase B. Activation of glutathione C. Inhibition of ALA synthase D. Inactivation of uroporphyrinogen III cosynthase

Correct Answer: A. Inhibition of ferrochelatase

Question 74

Clinical Scenario:

A 25-year-old woman presents to an urgent care center following a bee sting while at a picnic with her friends. She immediately developed a skin rash and swelling over her arm and face. She endorses diffuse itching over her torso. Past medical history is significant for a mild allergy to pet dander and ragweed. She occasionally takes oral contraceptive pills and diphenhydramine for her allergies. Family history is noncontributory. Her blood pressure is 119/81 mm Hg, heart rate is 101/min, respiratory rate is 21/min, and temperature is 37°C (98.6°F). On physical examination, the patient has severe edema over her face and severe stridor with inspiration at the base of both lungs. Of the following options, this patient is likely experiencing which of the following hypersensitivity reactions?

Answer Choices:

A. Type 1 - anaphylactic hypersensitivity reaction B. Type 2 - cytotoxic hypersensitivity reaction C. Type 4 - cell mediated (delayed) hypersensitivity reaction D. Both A & B

Correct Answer: A. Type 1 - anaphylactic hypersensitivity reaction

Question 75

Clinical Scenario:

A group of researchers wish to develop a clinical trial assessing the efficacy of a specific medication on the urinary excretion of amphetamines in intoxicated patients. They recruit 50 patients for the treatment arm and 50

patients for the control arm of the study. Demographics are fairly balanced between the two groups. The primary end points include (1) time to recovery of mental status, (2) baseline heart rate, (3) urinary pH, and (4) specific gravity. Which medication should they use in order to achieve a statistically significant result positively favoring the intervention?

Answer Choices:

A. Potassium citrate B. Ascorbic acid C. Aluminum hydroxide D. Tap water

Correct Answer: B. Ascorbic acid

Question 76

Clinical Scenario:

An epidemiologist is evaluating the efficacy of Noxbinle in preventing HCC deaths at the population level. He examines the survival data featured in the Noxbinle advertisement and wants to estimate how likely it is that Noxbinle will help an individual HCC patient. Based on the information provided in the drug advertisement, how many patients need to be treated with Noxbinle 100 mg in order to prevent, on average, one death from HCC?

Answer Choices:

A. 50 B. 20 C. 100 D. 10

Correct Answer: D. 10

Question 77

Clinical Scenario:

A 42-year-old woman comes to the physician for the evaluation of a 1-month history of dull lower abdominal pain, decreased appetite, and a 5-kg (11-lb) weight loss. Physical examination shows no abnormalities. Pelvic ultrasonography shows bilateral ovarian enlargement and free fluid in the rectouterine pouch. Biopsy specimens from the ovaries show multiple, round, mucin-filled cells with flat, peripheral nuclei. Further evaluation of this patient is most likely to show which of the following findings?

Answer Choices:

A. Decreased TSH levels B. Increased testosterone levels C. Dark blue peritoneal spots D. Gastric wall thickening

Correct Answer: D. Gastric wall thickening

Question 78

Clinical Scenario:

A 17-year-old girl is brought into the clinic by her mother who is concerned that she may be depressed. The mother states that her daughter feels unattractive and does not fit into any of the social groups at school. When talking to the patient, it is discovered that she mostly avoids the kids in school because of fear of rejection. She usually keeps to herself and says she hasn't involved herself in any group activities since elementary school. The patient's mother is worried that this kind of behavior might continue or worsen if it progresses into her college years. Which of the following is the most likely diagnosis in this patient?

Answer Choices:

A. Avoidant personality disorder B. Schizoid personality disorder C. Social phobia D. Body dysmorphic disorder

Correct Answer: A. Avoidant personality disorder

Question 79

Clinical Scenario:

On a medical trip to Nicaragua, you observe a sweet odor in the cerumen of 12-hour female newborn. Within 48 hours, the newborn develops ketonuria, poor feeding, and a sweet odor is also noticed in the urine. By 96 hours, the newborn is extremely lethargic and opisthotonus is observed. In order to prevent a coma and subsequent death, which of the following amino acids should be withheld from this newborn's diet?

Answer Choices:

A. Phenylalanine B. Valine C. Tyrosine D. Methionine

Correct Answer: B. Valine

Question 80

Clinical Scenario:

A 32-year-old HIV positive female known to be non-adherent to her treatment regimen, presents to the hospital with the complaint of new-onset headaches. Her vital signs are only significant for a low-grade fever. Neurological examination reveals right-sided upper motor neuron signs, as well as inattention and difficulty with concentration. The patient currently does not have a primary medical provider. A CT of the patient's head is shown in the image below. What is the next best step in management for this patient?

Answer Choices:

A. Perform a biopsy of the lesion B. Perform an analysis for 14-3-3 protein levels C. Begin treatment with pyrimethamine-sulfadiazine D. Begin treatment with acyclovir

Correct Answer: C. Begin treatment with pyrimethamine-sulfadiazine

Question 81

Clinical Scenario:

A 49-year-old man with hypertension comes to the hospital because of 4 days of left-sided chest pain, cough, and fever. The chest pain worsens upon inspiration and coughing. His temperature is 38.5°C (101.3° F), pulse is 110/min, respirations are 29/min. There is dullness to percussion at the left base of the lung. An x-ray of the chest shows blunting of the left costophrenic angle. Evaluation of the pleural fluid is most likely to show which of the following findings?

Answer Choices:

A. Increased lymphocyte concentration B. Decreased glucose concentration C. Increased pH D. Increased triglyceride concentration

Correct Answer: B. Decreased glucose concentration

Question 82

Clinical Scenario:

A 62-year-old man comes to the physician because of painless swelling in his left foot for 4 months. The swelling was initially accompanied by redness, which has since resolved. He has not had fever or chills. He has a history of coronary artery disease, hyperlipidemia, and type 2 diabetes mellitus. He has had 3 sexual partners over the past year and uses condoms inconsistently. His mother had rheumatoid arthritis. Current medications include clopidogrel, aspirin, metoprolol, losartan, atorvastatin, and insulin. He is 180 cm (5 ft 11 in) tall and weighs 95 kg (209 lb); BMI is 29 kg/m². Vital signs are within normal limits. Cardiovascular examination shows no abnormalities. Examination of the feet shows swelling of the left ankle with collapse of the midfoot arch and prominent malleoli. There is no redness or warmth. There is a small, dry ulcer on the left plantar surface of the 2nd metatarsal. Monofilament testing shows decreased sensation along both feet up to the shins bilaterally. His gait is normal. Which of the following is the most likely diagnosis?

Answer Choices:

A. Calcium pyrophosphate arthropathy B. Tertiary syphilis C. Reactive arthritis D. Diabetic arthropathy

Correct Answer: D. Diabetic arthropathy

Question 83

Clinical Scenario:

A 10-year-old boy presents to your office with puffy eyes. The patient's mother states that his eyes seem abnormally puffy and thinks he may have an eye infection. Additionally, he had a sore throat a week ago which resolved with over the counter medications. The mother also thought that his urine was darker than usual and is concerned that blood may be present. His temperature is 99.5°F (37.5°C), blood pressure is 107/62 mmHg, pulse is 100/min, respirations are 17/min, and oxygen saturation is 98% on room air. Physical exam is notable

for bilateral periorbital edema. Cranial nerves are grossly intact bilaterally. Which of the following is the most likely finding on renal biopsy for this patient

Answer Choices:

A. Linear Ig deposits along the basement membrane B. No abnormalities C. Podocyte fusion on electron microscopy D. Sub-epithelial electron dense deposits on electron microscopy

Correct Answer: D. Sub-epithelial electron dense deposits on electron microscopy

Question 84

Clinical Scenario:

A 55-year-old female comes to see her doctor for a rapidly enlarging abdomen and swelling of both legs. She was hospitalized 2 weeks ago for an upper gastrointestinal bleeding episode. On that occasion, her esophagogastroduodenoscopy showed grade 1 esophageal varices and a 1 cm clean-based antral ulcer. The patient was discharged on omeprazole. Review of symptoms shows that the patient is forgetful, does not sleep well, and is drowsy and fatigued during the day which prevents her from working full-time. She denies abdominal pain. The patient has a 10-year history of type 2 diabetes mellitus, hypertension, and hypercholesterolemia but no history of angina or coronary heart disease. She drank alcohol moderately heavy in her twenties and currently drinks less than 3 drinks per week and does not smoke. Her family history is unremarkable. On physical examination, her blood pressure is 132/82 mm Hg, pulse is 88/min, and her temperature is 37.0°C (98.6°F). She weighs 106.6 kg (235 lb) and her BMI is 33. She is alert, oriented to person, place, year, and month but not to the day. Her sclerae are nonicteric. Her pulmonary and cardiovascular exam are normal but her abdomen is distended with a fluid wave and mild tenderness to palpation. There is no hepatosplenomegaly. There is a 2+ edema to mid-calf and pedal pulses are barely palpable. Her neurological exam is without motor or sensory deficits but she demonstrates flapping tremor of her hands while asked to hold them in front of her for a few seconds and her skin exam shows a few spider telangiectasias on her face and upper chest. After an initial evaluation, lab tests were obtained: Serum sodium 133 mEq/L Serum potassium 3.8 mEq/L BUN 8 mg/dL Serum creatinine 1.0 mg/dL Serum albumin 2.5 mg/dL Aspartate aminotransferase 68 IU/ml Alanine aminotransferase 46 IU/ml Alkaline phosphatase 130 IU/ml Total bilirubin 1.8 mg/dL WBC count 4,200/mm³ Platelets 94,000/mm³ Hematocrit 35.5% Prothrombin time (INR) 1.5 A liver biopsy is performed and the results are pending. The hepatocytes causing her acute issue are predominantly located in which area of the hepatic lobule?

Answer Choices:

A. The zone where gluconeogenesis is predominant B. The zone with little or no cytochrome P450 enzymes C. The zone closest to the centrolobular vein D. The zone involved in cholesterol synthesis

Correct Answer: C. The zone closest to the centrolobular vein

Question 85

Clinical Scenario:

Three days after undergoing coronary artery bypass surgery, a 72-year-old man has severe right upper quadrant pain, fever, nausea, and vomiting. He has type 2 diabetes mellitus, benign prostatic hyperplasia, peripheral vascular disease, and chronic mesenteric ischemia. He had smoked one pack of cigarettes daily for 30 years but quit 10 years ago. He drinks 8 cans of beer a week. His preoperative medications include metformin, aspirin, simvastatin, and finasteride. His temperature is 38.9°C (102°F), pulse is 102/min, respirations are 18/min, and blood pressure is 110/60 mmHg. Auscultation of the lungs shows bilateral inspiratory crackles. Cardiac examination shows no murmurs, rubs or gallops. Abdominal examination shows soft abdomen with tenderness and sudden inspiratory arrest upon palpation in the right upper quadrant. There is no rebound tenderness or guarding. Laboratory studies show the following: Hemoglobin 13.1 g/dL Hematocrit 42% Leukocyte count 15,700/mm³ Segmented neutrophils 65% Bands 10% Lymphocytes 20% Monocytes 3% Eosinophils 1% Basophils 0.5% AST 40 U/L ALT 100 U/L Alkaline phosphatase 85 U/L Total bilirubin 1.5 mg/dL Direct 0.9 mg/dL Amylase 90 U/L Abdominal ultrasonography shows a distended gallbladder, thickened gallbladder wall with pericholecystic fluid, and no stones. Which of the following is the most appropriate next step in management?"

Answer Choices:

A. Intravenous heparin therapy followed by embolectomy B. Careful observation with serial abdominal examinations C. Immediate cholecystectomy D. Intravenous piperacillin-tazobactam therapy and percutaneous cholecystostomy

Correct Answer: D. Intravenous piperacillin-tazobactam therapy and percutaneous cholecystostomy

Question 86

Clinical Scenario:

An investigator is studying metabolic processes in cells from a mouse model. She identifies certain cells that are unable to generate enough reducing factor for respiratory burst. Increased production of which of the following substances is most likely to be present in these cells?

Answer Choices:

A. Ribulose-5-phosphate from glucose-6-phosphate B. Mevalonate from β -hydroxy- β -methylglutaryl-CoA C. Ribose-5-phosphate from fructose-6-phosphate D. 6-phosphogluconolactone from glucose-6-phosphate

Correct Answer: C. Ribose-5-phosphate from fructose-6-phosphate

Question 87

Clinical Scenario:

A 70-year-old woman with history of coronary artery disease status-post coronary artery bypass graft presents with a stroke due to an infarction in the right middle cerebral artery territory. She is admitted to the intensive care unit for neurological monitoring following a successful thrombectomy. Overnight, the patient complains of difficulty breathing, chest pain, and jaw pain. Her temperature is 98.6°F (37°C), blood pressure is 160/80 mmHg, pulse is 100/min, respirations are 30/min, and oxygen saturation is 90% on 2L O₂ via nasal cannula.

Rales are heard in the lower lung bases. Electrocardiogram reveals left ventricular hypertrophy with repolarization but no acute ST or T wave changes. Troponin is 2.8 ng/mL. Chest radiograph reveals Kerley B lines. After administration of oxygen, aspirin, carvedilol, and furosemide, the patient improves. The next troponin is 3.9 ng/mL. Upon further discussion with the consulting cardiologist and neurologist, a heparin infusion is started. After transfer to a general medicine ward floor four days later, the patient complains of a headache. The patient's laboratory results are notable for the following:

Hemoglobin: 11 g/dL Hematocrit: 36% Leukocyte count: 11,000 /mm³ with normal differential Platelet count: 130,000 /mm³

On admission, the patient's platelet count was 300,000/mm³. What medication is appropriate at this time?

Answer Choices:

A. Argatroban B. Dalteparin C. Protamine D. Tinzaparin

Correct Answer: A. Argatroban

Question 88

Clinical Scenario:

A 5-year-old boy is brought in by his mother with reports of trouble at school. Teachers report that for the last 6 months he has been having difficulty finishing tasks, is easily distracted, frequently does not listen, commonly fails to finish schoolwork, has not been able to complete any of the class projects this year, and frequently loses school books and supplies. Teachers also say that he constantly fidgets, often leaves his seat without permission, has trouble being quiet, talks excessively, frequently interrupts his classmates when trying to answer questions, and has difficulty waiting in line. The mother states that she has also been noticing similar behaviors at home and that his symptoms have been affecting him negatively academically and socially. The patient has no significant past medical history. The patient is in the 90th percentile for height and weight and has been meeting all the developmental milestones. He is afebrile, and his vital signs are within normal limits. A physical examination is unremarkable. Which of the following medications is a first-line treatment for this patient's most likely diagnosis?

Answer Choices:

A. Atomoxetine B. Methylphenidate C. Clonidine D. Guanfacine

Correct Answer: B. Methylphenidate

Question 89

Clinical Scenario:

A group of scientists discovered a neurotoxin that prevents neurons from releasing neurotransmitters. They performed a series of experiments to determine the protein that the neurotoxin affected. They used a fluorescent molecule that localizes to synaptic vesicles. In the control experiment, they observed the movement of vesicles from the cell body down the axon and finally to the synapse, and they saw movement from the synapse back to

the cell body. When the neurotoxin was applied, the vesicles stopped moving down the axon, but movement back to the cell body still occurred. They also applied tetanospasmin and botulinum toxin to see if these toxins exhibited similar behavior but they did not. Which of the following proteins is most likely affected by this neurotoxin?

Answer Choices:

A. Kinesin B. Dynein C. Synaptobrevin D. Alpha/Beta tubulin

Correct Answer: A. Kinesin

Question 90

Clinical Scenario:

A 68-year-old man presents to your office concerned about the frequency with which he wakes up in the middle night feeling out of breath. He has been required to use 3 more pillows in order to reduce the shortness of breath. In addition to this, he has noticed that he tires easily with minor tasks such as walking 1 block, and more recently when he is dressing up in the mornings. Physical examination reveals a blood pressure of 120/85 mm Hg, heart rate of 82/min, respiratory rate of 20/min, and body temperature of 36.0°C (98.0°F).

Cardiopulmonary examination reveals regular and rhythmic heart sounds with S4 gallop, a laterally displaced point of maximum impulse (PMI), and rales in both lung bases. He also presents with prominent hepatjugular reflux, orthopnea, and severe lower limb edema. Which of the following changes would be seen in this patient's heart?

Answer Choices:

A. Increased nitric oxide bioactivity B. Decreased collagen synthesis C. Increased production of brain natriuretic peptide D. Cardiomyocyte hyperplasia

Correct Answer: C. Increased production of brain natriuretic peptide

Question 91

Clinical Scenario:

A 33-year-old man is brought to the emergency department by his mother because of erratic behavior over the past 6 months. He spends most of his time alone in his room because he believes he is being followed by the Secret Service. He was fired from his job 3 months ago after threatening a colleague. He appears suspicious of his surroundings and asks the doctor questions about "the security of the hospital." The patient exhibits a flat affect. During physical examination, he tells the doctor that he has a gun at home and plans to shoot his neighbor, whom he thinks is working for the Secret Service. Which of the following is the most appropriate action by the doctor?

Answer Choices:

A. Discuss the diagnosis and therapy plan with the patient's mother B. Inform security and law enforcement C. Warn the person at risk and inform law enforcement D. Request a court order to override the patient's right to confidentiality

Correct Answer: C. Warn the person at risk and inform law enforcement

Question 92

Clinical Scenario:

A 72-year-old multiparous woman comes to the physician for the evaluation of episodes of involuntary urine leakage for the past 6 months. She loses small amounts of urine without warning after laughing or sneezing. She also sometimes forgets the names of her grandchildren and friends. She is retired and lives at an assisted-living facility. She has insulin-dependent diabetes mellitus type 2. Her mother received a ventriculoperitoneal shunt around her age. She walks without any problems. Sensation to pinprick and light touch is normal. Which of the following is the primary underlying etiology for this patient's urinary incontinence?

Answer Choices:

A. Detrusor-sphincter dyssynergia B. Urethral hypermobility C. Impaired detrusor contractility D. Loss of sphincter control

Correct Answer: B. Urethral hypermobility

Question 93

Clinical Scenario:

A 38-year-old woman comes to the physician because of an 8-week history of shortness of breath and dull chest pain. She has a history of antiphospholipid syndrome. Physical examination shows jugular venous distention. Right heart catheterization shows a mean pulmonary arterial pressure of 30 mm Hg and a pulmonary capillary wedge pressure of 10 mm Hg. Further evaluation is most likely to show which of the following?

Answer Choices:

A. Constriction of the renal afferent arteriole B. Dilation of the coronary sinus C. Decreased left ventricular contractility D. Hemosiderin-laden macrophages "

Correct Answer: B. Dilation of the coronary sinus

Question 94

Clinical Scenario:

A 31-year-old male presents to the emergency room following an altercation with patrons at a local grocery store. He is acting aggressively toward hospital staff and appears to be speaking to non-existent individuals. On

examination he is tachycardic and diaphoretic. Horizontal and vertical nystagmus is noted. The patient eventually admits to taking an illegal substance earlier in the evening. Which of the following mechanisms of action is most consistent with the substance this patient took?

Answer Choices:

A. Adenosine antagonist B. Mu receptor agonist C. GABA agonist D. NMDA receptor antagonist

Correct Answer: D. NMDA receptor antagonist

Question 95

Clinical Scenario:

A 55-year-old man comes to the physician for a follow-up examination. For the past 6 months, he has had fatigue, headaches, and several episodes of dizziness. Three months ago, he was diagnosed with hypertension and started on medications. Since the diagnosis was made, his medications have been adjusted several times because of persistently high blood pressure readings. He also has hypercholesterolemia and peripheral arterial disease. He smoked one pack of cigarettes daily for 34 years but quit two months ago. His current medications include aspirin, atorvastatin, losartan, felodipine, and hydrochlorothiazide. He is 188 cm (6 ft 2 in) tall and weighs 109 kg (240 lb); BMI is 31 kg/m². His pulse is 82/min and blood pressure is 158/98 mm Hg. Physical examination shows bilateral carotid bruits and normal heart sounds. Serum potassium concentration is 3.2 mEq/L, plasma renin activity is 4.5 ng/mL/h (N = 0.3–4.2 ng/mL/h), and serum creatinine concentration is 1.5 mg/dL. Further evaluation of this patient is most likely to show which of the following findings?

Answer Choices:

A. Unilateral kidney atrophy B. Bilateral kidney enlargement C. Pituitary mass D. Diffuse thyroid enlargement

Correct Answer: A. Unilateral kidney atrophy

Question 96

Clinical Scenario:

A 30-year-old woman came to her OBGYN for an infertility consultation. The patient reports having intercourse with her husband at least 3 times per week with increasing frequency during the periods. The lab reports of her husband revealed an adequate sperm count. After the work-ups was complete, her OBGYN prescribed a medication similar to GnRH to be administered in a pulsatile manner. Which drug is prescribed to the patient?

Answer Choices:

A. Leuprolide B. Anastrozole C. Clomiphene D. Mestranol

Correct Answer: A. Leuprolide

Question 97

Clinical Scenario:

A 17-year-old girl is brought to the physician by her mother for the evaluation of irregular menstrual bleeding. Menses have occurred at 60- to 90-day intervals since menarche at the age of 12 years. Her last menstrual period was 4 weeks ago. She is sexually active with one male partner, and they use condoms consistently. She reports that she currently has no desire to have children. She is 165 cm (5 ft 5 in) tall and weighs 85 kg (187 lb); BMI is 31 kg/m². Examination shows scattered pustules on the forehead and oily skin. There is coarse hair on the chin and upper lip. Fingerstick blood glucose concentration is 190 mg/dL. A urine pregnancy test is negative. Which of the following is the most appropriate pharmacotherapy?

Answer Choices:

A. Danazol B. Leuprolide C. Metformin D. Combination oral contraceptives

Correct Answer: D. Combination oral contraceptives

Question 98

Clinical Scenario:

A 35-year-old man comes to the physician because of dull abdominal pain on his right side for 4 months. He also reports episodic nausea and vomiting during this period. He does not have fever, altered bowel habits, or weight loss. He has had a pet dog for 8 years. He appears healthy. Vital signs are within normal limits. Abdominal examination shows a nontender mass 3 cm below the right costal margin that moves with respiration. Laboratory studies show: Hemoglobin 14.6 g/dL Leukocyte count 7200/mm³ Segmented neutrophils 58% Eosinophils 8% Lymphocytes 30% Monocytes 4% Ultrasound of the abdomen shows a focal hypoechoic cyst within the liver measuring 7 cm. An ELISA confirms the diagnosis. He is scheduled for CT-guided percutaneous drainage under general anesthesia with orotracheal intubation. Seven minutes into the procedure, the patient's oxygen saturation suddenly decreases from 95% to 64%. His heart rate is 136/min, and blood pressure is 86/58 mm Hg. Capnography cannot record an end tidal CO₂ waveform. Breath sounds are absent bilaterally. Which of the following is most appropriate next step in management?"

Answer Choices:

A. Exploratory laparotomy B. Epinephrine C. Cricothyrotomy D. Norepinephrine "

Correct Answer: B. Epinephrine

Question 99

Clinical Scenario:

A 10-month-old boy is brought to his pediatrician because of a 3-day history of fever and lethargy. He has previously had more infections than expected since birth but otherwise appears to be developing normally. On exam, the boy is found to have a purulent, erythematous bump on his left upper extremity. This lesion is

cultured and found to have a catalase-positive, coagulase-positive, gram-positive organism, which is the same organism that caused his previous infections. Based on clinical suspicion, an incubated leukocyte test is obtained that confirms the diagnosis. The substrate of the protein that is most likely defective in this patient is produced by which of the following metabolic pathways?

Answer Choices:

A. Beta oxidation B. Citric acid cycle C. Gluconeogenesis D. HMP shunt

Correct Answer: D. HMP shunt

Question 100

Clinical Scenario:

A 36-year-old woman presents with a whitish vaginal discharge over the last week. She also complains of itching and discomfort around her genitals. She says her symptoms are getting progressively worse. She has been changing her undergarments frequently and changed the brand of detergent she uses to wash her clothes, but it did not resolve her problem. Additionally, she admits to having painful urination and increased urinary frequency for the past one month, which she was told are expected side effects of her medication. The patient denies any recent history of fever or malaise. She has 2 children, both delivered via cesarean section in her late twenties. Past medical history is significant for hypertension and diabetes mellitus type 2. Current medications are atorvastatin, captopril, metformin, and empagliflozin. Her medications were changed one month ago to improve her glycemic control, as her HbA1c at that time was 7.5%. Her vital signs are a blood pressure of 126/84 mm Hg and a pulse of 78/min. Her fingerstick glucose is 108 mg/dL. Pelvic examination reveals erythema and mild edema of the vulva. A thick, white, clumpy vaginal discharge is seen. The vaginal pH is 4.0. Microscopic examination of a KOH-treated sample of the discharge demonstrates lysis of normal cellular elements with branching pseudohyphae. Which of the following is the next best step in the management of this patient?

Answer Choices:

A. Stop empagliflozin. B. Advise her to drink lots of cranberry juice. C. Switch her from oral antidiabetic medication to insulin. D. Start fluconazole.

Correct Answer: D. Start fluconazole.

Question 101

Clinical Scenario:

A 60-year-old man comes to the physician because of persistent fatigue over the past ten months. His previous annual health maintenance examination showed no abnormalities. He appears pale. Physical examination shows numerous petechial lesions over the abdomen and marked splenomegaly. His serum hemoglobin concentration is 9.4 g/dL, leukocyte count is 4,100/mm³, and thrombocyte count is 110,000/mm³. A peripheral blood smear shows large white blood cells with centrally placed nuclei and multiple fine, radial cytoplasmic

projections that stain positively for tartrate-resistant acid phosphatase (TRAP). Which of the following is the most likely characteristic of the medication used as first-line treatment for this patient's condition?

Answer Choices:

A. Increases risk of thromboembolic events B. Resistant to breakdown by adenosine deaminase C. Requires bioactivation by the liver D. Unable to cross the blood-brain barrier

Correct Answer: B. Resistant to breakdown by adenosine deaminase

Question 102

Clinical Scenario:

A physician is choosing whether to prescribe losartan or lisinopril to treat hypertension in a 56-year-old male. Relative to losartan, one would expect treatment with lisinopril to produce which of the following changes in the circulating levels of these peptides?

Answer Choices:

A. Bradykinin increase; angiotensin II decrease B. Renin decrease; angiotensin 1 increase C. Aldosterone increase; bradykinin decrease D. Renin decrease; angiotensin II increase

Correct Answer: A. Bradykinin increase; angiotensin II decrease

Question 103

Clinical Scenario:

An 11-month-old male is brought to the emergency room by his mother. The mother reports that the child is in severe pain and has not moved his right leg since earlier this morning when he was crawling on the floor. The child did not fall or sustain any obvious injury. The child's past medical history is notable for anemia and recurrent infections since birth. His temperature is 99.1°F (37.3°C), blood pressure is 100/65 mmHg, pulse is 120/min, and respirations are 22/min. Physical examination reveals mild macrocephaly and hepatosplenomegaly. Palpation of the right femur seems to exacerbate the child's pain. A radiograph demonstrates a transverse mid-shaft femur fracture. Which of the following is the most likely cause of this patient's condition?

Answer Choices:

A. Acidification deficiency B. Vitamin deficiency C. Surreptitious child abuse D. Collagen production deficiency

Correct Answer: A. Acidification deficiency

Question 104

Clinical Scenario:

A 43-year-old woman comes to the physician because of tingling and weakness in her left arm for the past 2 days. An image of the brachial plexus is shown. Nerve conduction study shows decreased transmission of electrical impulses in the labeled structure. Physical examination is most likely to show impairment of which of the following movements?

Answer Choices:

- A. Extension of the wrist and fingers
- B. Opposition of the thumb
- C. Flexion of the metacarpophalangeal joints
- D. Abduction of the shoulder above 100 degrees

Correct Answer: A. Extension of the wrist and fingers

Question 105

Clinical Scenario:

A 22-year-old man comes to the physician because of a fall associated with a 6-month history of increasing difficulty walking. Over the last year, his friends have also noticed his speech becoming slower. During this period, he also gave up his hobby of playing video games because he has become clumsy with his hands. His father died of esophageal varices at the age of 40 years. The patient does not smoke or drink alcohol. He takes no medications. He appears sad. His temperature is 37°C (98.6°F), pulse is 70/min, and blood pressure is 120/80 mm Hg. He is alert and oriented to person, place, and time. His speech is slurred and monotonous; his gait is unsteady. Examination shows scleral icterus and some drooling. The liver is palpated 2 to 3 cm below the right costal margin, and the spleen is palpated 1 to 2 cm below the left costal margin. Further evaluation of this patient is most likely to show which of the following findings?

Answer Choices:

- A. Increased number of CAG repeats
- B. Oligoclonal bands on CSF analysis
- C. Ventriculomegaly on CT scan of the brain
- D. Low serum ceruloplasmin concentration

Correct Answer: D. Low serum ceruloplasmin concentration

Question 106

Clinical Scenario:

A 32-year-old man is brought to the emergency department because of a 2-day history of confusion and rapidly progressive dyspnea. He has had a fever and chills for the past five days. Five years ago, he was diagnosed with hepatitis C. He has smoked two packs of cigarettes daily for 15 years and drinks one to two beers daily. He has a history of past intravenous heroin use. He appears pale, anxious, and in severe distress. His temperature is 39.3°C (102.7°F), respirations are 30/min, pulse is 59/min, and blood pressure is 80/50 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 75%. Examination shows multiple linear hemorrhages underneath the nails. There are nontender maculae on both palms and soles. Fine rales are heard bilaterally on auscultation of the chest. Cardiac examination shows an S3; a grade 3/6 high-pitched decrescendo early diastolic murmur is heard along the left sternal border and right second intercostal space. An x-ray of the chest shows a normal sized heart and pulmonary edema. An ECG shows P waves and QRS

complexes that occur at regular intervals, but independently of each other. A transesophageal echocardiography (TEE) is most likely to show which of the following?

Answer Choices:

A. A highly echogenic, thin, linear structure in the right atrium B. Anechoic space between pericardium and epicardium C. Perivalvular thickening with an echolucent cavity D. Oscillating mobile mass on the tricuspid valve

Correct Answer: C. Perivalvular thickening with an echolucent cavity

Question 107

Clinical Scenario:

A 2-year-old girl is brought to the emergency department because of bilateral hand pain and swelling. Her parents say the pain began 1 week ago and has gotten progressively worse. Two weeks ago, the patient had a low-grade fever and lace-like rash on her arms and trunk for several days. The patient appears to be in distress. Her temperature is 38.5°C (101.4°F), pulse is 130/min, and respirations are 25/min. The dorsum of her hands and fingers are erythematous, swollen, warm, and tender to palpation. Her hemoglobin concentration is 9.1 g/dL and leukocyte count is 8,000/mm³. A peripheral blood smear is shown. Which of the following interventions is most appropriate to prevent a recurrence of this patient's symptoms?

Answer Choices:

A. Hydroxyurea B. Regular red cell exchange transfusions C. Prophylactic penicillin D. IV cefazolin

Correct Answer: A. Hydroxyurea

Question 108

Clinical Scenario:

A 24-year-old woman presents with a 3-month history of bloody diarrhea and intermittent abdominal pain. She says that after she has a bowel movement, she still feels as though she needs to go more. She also reports a 10-pound weight loss, significant fatigue, and frequent cravings to chew ice. Her past medical history is significant only for chronic iron deficiency anemia since high school. She currently takes a women's multivitamin and ferrous sulfate 65 mg orally once daily. She is on the college track team but now is too tired to participate in practice. Her family history is significant for colon cancer and her grandmother died from breast cancer in her 70's. Her vital signs include: temperature 37.0°C (98.6°F), pulse 102/min, respiratory rate 16/min, blood pressure 100/75 mm Hg. Physical examination is significant for conjunctival pallor, koilonychia, and the cutaneous findings shown in the exhibit. Laboratory tests show elevated ESR and C-reactive protein and findings consistent with iron deficiency anemia. A barium enema demonstrates a lead pipe appearance and a loss of haustra. Which of the following are the recommended screening guidelines for colorectal cancer for this patient?

Answer Choices:

A. Colonoscopy by age 32 and repeated every 1–2 years thereafter B. Colonoscopy by age 50 and repeated every 10 years thereafter C. Colonoscopy by age 32 and repeated every 3 years thereafter D. Flexible sigmoidoscopy by age 32 and repeated every 1–2 years thereafter

Correct Answer: A. Colonoscopy by age 32 and repeated every 1–2 years thereafter

Question 109

Clinical Scenario:

An 84-year-old man comes to the emergency department because of lower back pain and lower extremity weakness for 3 weeks. Over the past week, he has also found it increasingly difficult to urinate. He has a history of prostate cancer, for which he underwent radical prostatectomy 8 years ago. His prostate-specific antigen (PSA) level was undetectable until a routine follow-up visit last year, when it began to increase from 0.8 ng/mL to its present value of 64.3 ng/mL ($N < 4$). An MRI of the spine shows infiltrative vertebral lesions with a collapse of the L5 vertebral body, resulting in cord compression at L4–L5. The patient receives one dose of intravenous dexamethasone and subsequently undergoes external beam radiation. Which of the following cellular changes is most likely to occur as a result of this treatment?

Answer Choices:

A. Formation of pyrimidine dimers B. Intercalation of neighbouring DNA base pairs C. Generation of hydroxyl radicals D. Formation of DNA crosslinks

Correct Answer: C. Generation of hydroxyl radicals

Question 110

Clinical Scenario:

An otherwise healthy 7-year-old boy is brought to the emergency department because of a 1-day history of involuntary muscle contractions and pain in his back and neck. Two weeks ago, he fell while playing in the sandbox and scraped both his knees. He has not received any vaccinations since birth. His temperature is 38.5°C (101.3°F). He is diaphoretic. Examination shows inability to open his mouth beyond 1 cm. There is hyperextension of the lumbar spine and resistance to neck flexion. Administration of which of the following would most likely have prevented this patient's current condition?

Answer Choices:

A. Capsular polysaccharides B. Chemically-inactivated virus C. Denaturated bacterial product D. Viable but weakened microorganism

Correct Answer: C. Denaturated bacterial product

Question 111

Clinical Scenario:

An investigator inoculates three different broths with one colony-forming unit of *Escherichia coli*. Broth A contains 100 μmol of lactose, broth B contains 100 μmol of glucose, and broth C contains both 100 μmol of lactose and 100 μmol of glucose. After 24 hours, the amounts of lactose, galactose, and glucose in the three broths are measured. The results of the experiment are shown:

	Lactose	Galactose	Glucose
Broth A	43 μmol	11 μmol	9 μmol
Broth B	0 μmol	0 μmol	39 μmol
Broth C	94 μmol	1 μmol	66 μmol

The observed results are most likely due to which of the following properties of broth A compared to broth C?"

Answer Choices:

A. Increased activity of glycosylases
B. Decreased activity of catabolite activator protein
C. Decreased production of α -galactosidase
D. Increased activity of adenylate cyclase

Correct Answer: D. Increased activity of adenylate cyclase

Question 112

Clinical Scenario:

A 15-year-old boy is brought to the physician because of recurrent respiratory infections that cause him to miss several weeks of school each year. He also has bulky, foul-smelling stools that are difficult to flush. He has a good appetite and eats a variety of foods. His height and weight are below the 10th percentile. Physical examination shows multiple nasal polyps. There is mild wheezing over the lower lung fields. Further evaluation is most likely to show which of the following?

Answer Choices:

A. Antibodies to endomysium
B. Absence of the vas deferens
C. Positive methacholine challenge test
D. Deficiency of immunoglobulin A

Correct Answer: B. Absence of the vas deferens

Question 113

Clinical Scenario:

A 2-week-old infant is brought to the physician by her father because of a 1-week history of vaginal discharge. The discharge was initially clear, but now he notices that it is tinged with blood. The father is also concerned about "bruises" on his daughter's back and buttocks. Both parents work so that the infant spends most of her time in daycare or with her aunt. She was born at term following a pregnancy complicated by maternal gonococcal infection that was treated with antibiotics. She appears well. Physical examination shows mild acne across her cheeks and forehead. There are multiple large flat gray-blue patches on her back and buttocks. An image of one of the lesions is shown. Firm breast buds are present. Genitourinary examination shows erythema

and swelling of the vulva and vagina with an odorless, blood-stained white discharge. Which of the following is the most appropriate next step in management?

Answer Choices:

A. Reassurance B. Ceftriaxone and doxycycline therapy C. Leuprolide therapy D. Fluconazole therapy

Correct Answer: A. Reassurance

Question 114

Clinical Scenario:

A 38-year-old woman comes to the physician for a follow-up examination. Two years ago, she was diagnosed with multiple sclerosis. Three weeks ago, she was admitted and treated for right lower leg weakness with high-dose methylprednisone for 5 days. She has had 4 exacerbations over the past 6 months. Current medications include interferon beta and a multivitamin. Her temperature is 37°C (98.6°F), pulse is 90/min, and blood pressure is 116/74 mm Hg. Examination shows pallor of the right optic disk. Neurologic examination shows no focal findings. She is anxious about the number of exacerbations and repeated hospitalizations. She is counseled about the second-line treatment options available to her. She consents to treatment with natalizumab. However, she has read online about its adverse effects and is concerned. This patient is at increased risk for which of the following complications?

Answer Choices:

A. Progressive multifocal leukoencephalopathy B. Parkinsonism C. Tuberculosis D. Aplastic anemia

Correct Answer: A. Progressive multifocal leukoencephalopathy

Question 115

Clinical Scenario:

A 42-year-old woman with a history of multiple sclerosis and recurrent urinary tract infections comes to the emergency department because of flank pain and fever. Her temperature is 38.8°C (101.8°F). Examination shows left-sided costovertebral angle tenderness. She is admitted to the hospital and started on intravenous vancomycin. Three days later, her symptoms have not improved. Urine culture shows growth of *Enterococcus faecalis*. Which of the following best describes the most likely mechanism of antibiotic resistance in this patient?

Answer Choices:

A. Production of beta-lactamase B. Alteration of penicillin-binding proteins C. Alteration of peptidoglycan synthesis D. Alteration of ribosomal targets

Correct Answer: C. Alteration of peptidoglycan synthesis

Question 116

Clinical Scenario:

A 53-year-old man comes to the physician because of a 3-month history of a nonpruritic rash, fatigue, and decreased urination. Physical examination shows multiple erythematous, purpuric papules on his trunk and extremities that do not blanch when pressed. Serum creatinine is elevated and urinalysis shows red blood cell casts and protein. Serum complement levels are decreased. Renal biopsy shows subendothelial immune complex deposits with granular immunofluorescence and tram-track basement membrane splitting. Further laboratory evaluation of this patient is most likely to show the presence of which of the following antibodies?

Answer Choices:

A. Anti-desmoglein antibodies B. Anti-hepatitis C antibodies C. Anti-DNA topoisomerase antibodies D. Anticardiolipin antibodies

Correct Answer: B. Anti-hepatitis C antibodies

Question 117

Clinical Scenario:

A 5-year-old boy presents to his pediatrician with weakness. His father observed that his son seemed less energetic at daycare and kindergarten classes. He was becoming easily fatigued from mild play. His temperature is 98°F (37°C), blood pressure is 90/60 mmHg, pulse is 100/min, and respirations are 20/min. Physical exam reveals pale conjunctiva, poor skin turgor and capillary refill, and cervical and axillary lymphadenopathy with assorted bruises throughout his body. A complete blood count reveals the following:

Leukocyte count: 3,000/mm³ Segmented neutrophils: 30% Bands: 5% Eosinophils: 5% Basophils: 10% Lymphocytes: 40% Monocytes: 10%

Hemoglobin: 7.1 g/dL Hematocrit: 22% Platelet count: 50,000/mm³

The most specific diagnostic assessment would most likely show which of the following?

Answer Choices:

A. Bone marrow biopsy with > 25% lymphoblasts B. Flow cytometry with positive terminal deoxynucleotidyl transferase staining C. Fluorescence in situ hybridization analysis with 9:22 translocation D. Peripheral blood smear with > 50% lymphoblasts

Correct Answer: A. Bone marrow biopsy with > 25% lymphoblasts

Question 118

Clinical Scenario:

A 4-year-old boy is brought to the emergency department with difficulty breathing. His mother reports that he developed a fever last night and began to have trouble breathing this morning. The boy was born at 39 weeks

gestation via spontaneous vaginal delivery. He is unvaccinated (conscientious objection by the family) and is meeting all developmental milestones. At the hospital, his vitals are temperature 39.8°C (103.6°F), pulse 122/min, respiration rate 33/min, blood pressure 110/66 mm Hg, and SpO2 93% on room air. On physical examination, he appears ill with his neck hyperextended and chin protruding. His voice is muffled and is drooling. The pediatrician explains that there is one particular bacteria that commonly causes these symptoms. At what age should the patient have first received vaccination to prevent this condition from this particular bacteria?

Answer Choices:

A. At 2-months-old B. At 6-months-old C. Between 9- and 12-months-old D. Between 12- and 15-months-old

Correct Answer: A. At 2-months-old

Question 119

Clinical Scenario:

A 2755-g (6-lb 1-oz) baby boy is delivered at 37 weeks' gestation to a 29-year-old woman who is gravida 3, para 3. His mother received no prenatal care during her pregnancy. 12 hours after birth, he is evaluated for jaundice and lethargy. Laboratory studies show a hemoglobin concentration of 9.6 g/dL and a serum total bilirubin concentration of 10 mg/dL. The results of a direct Coombs test are positive. Further evaluation is most likely to show which of the following?

Answer Choices:

A. Hyposthenuria B. Positive eosin-5-maleimide binding test C. Hepatosplenomegaly D. Elevated urinary coproporphyrins

Correct Answer: C. Hepatosplenomegaly

Question 120

Clinical Scenario:

A 44-year-old man presents to the clinic with recurrent epigastric pain following meals for a month. He adds that the pain radiates up his neck and throat. Over the counter antacids have not helped. On further questioning, he endorses foul breath upon waking in the morning and worsening of pain when lying down. He denies any recent weight loss. His temperature is 37°C (98.6°F), respirations are 15/min, pulse is 70/min, and blood pressure is 100/84 mm Hg. A physical examination is performed which is within normal limits except for mild tenderness on deep palpation of the epigastrium. An ECG performed in the clinic shows no abnormalities. What is the next best step in the management of this patient?

Answer Choices:

A. Barium swallow B. Lansoprazole C. Liquid antacid D. Ranitidine

Correct Answer: B. Lansoprazole

Question 121

Clinical Scenario:

A 4-year-old girl is brought to the physician because her mother is concerned that she has been talking to an imaginary friend for 2 months. The child calls her friend 'Lucy' and says "Lucy is my best friend". The child has multiple conversation and plays with the 'Lucy' throughout the day. The girl attends preschool regularly. She can copy a circle, tells stories, and can hop on one foot. Her maternal uncle has schizophrenia. Her parents are currently divorcing. The child's father has a history of illicit drug use. Physical examination shows no abnormalities. The mother is concerned about whether the child is acting out because of the divorce. Which of the following is the most appropriate next best step in management?

Answer Choices:

A. Perform MRI of the brain B. Inform Child Protective Services C. Schedule psychiatry consult D. Reassure the mother

Correct Answer: D. Reassure the mother

Question 122

Clinical Scenario:

A 30-year-old man presents with fever, malaise, and severe pain in his right wrist and left knee for the last 2 days. He describes the pain as 8/10 in intensity, sharp in character, and extending from his right wrist to his fingers. He denies any recent inciting trauma or similar symptoms in the past. His past medical history is unremarkable. He is sexually active with multiple partners and uses condoms inconsistently. The vital signs include blood pressure 120/70 mm Hg, pulse 100/min, and temperature 38.3°C (101.0°F). On physical examination, the right wrist and left knee joints are erythematous, warm, and extremely tender to palpation. Both joints have a significantly restricted range of motion. A petechial rash is noted on the right forearm. An arthrocentesis is performed on the left knee joint. Which of the following would be the most likely finding in this patient?

Answer Choices:

A. Arthrocentesis aspirate showing gram-positive cocci in clusters B. Arthrocentesis aspirate showing minimal, purulent joint effusion with negative culture C. Positive serum ASO titer D. Radiographs of right wrist and left knee showing osteopenia and joint space narrowing

Correct Answer: B. Arthrocentesis aspirate showing minimal, purulent joint effusion with negative culture

Question 123

Clinical Scenario:

Please refer to the summary above to answer this question Further evaluation of this patient is most likely to show which of the following findings?" "Patient Information Age: 28 years Gender: F, self-identified Ethnicity:

unspecified Site of Care: office History Reason for Visit/Chief Concern: "I'm not making breast milk anymore." History of Present Illness: 1-week history of failure to lactate; has previously been able to breastfeed her twins, who were born 12 months ago menses resumed 4 months ago but have been infrequent feels generally weak and tired has had a 6.8-kg (15-lb) weight gain over the past 2 months despite having a decreased appetite Past Medical History: vaginal delivery of twins 12 months ago, complicated by severe postpartum hemorrhage requiring multiple blood transfusions atopic dermatitis Social History: does not smoke, drink alcohol, or use illicit drugs is not sexually active Medications: topical triamcinolone, multivitamin Allergies: no known drug allergies Physical Examination Temp Pulse Resp BP O2 Sat Ht Wt BMI 37°C (98.6°F) 54/min 16/min 101/57 mm Hg – 160 cm (5 ft 3 in) 70 kg (154 lb) 27 kg/m2 Appearance: tired-appearing HEENT: soft, nontender thyroid gland without nodularity Pulmonary: clear to auscultation Cardiac: bradycardic but regular rhythm; normal S1 and S2; no murmurs, rubs, or gallops Breast: no nodules, masses, or tenderness; no nipple discharge Abdominal: overweight; no tenderness, guarding, masses, bruits, or hepatosplenomegaly; normal bowel sounds Extremities: mild edema of the ankles bilaterally Skin: diffusely dry Neurologic: alert and oriented; cranial nerves grossly intact; no focal neurologic deficits; prolonged relaxation phase of multiple deep tendon reflexes"

Answer Choices:

A. Increased serum sodium concentration B. Increased serum TSH concentration C. Decreased serum oxytocin concentration D. Decreased serum cortisol concentration

Correct Answer: C. Decreased serum oxytocin concentration

Question 124

Clinical Scenario:

A previously healthy 4-year-old boy is brought to the emergency department because of a 1-day history of pain and swelling of his left knee joint. He has not had any trauma to the knee. His family history is unremarkable except for a bleeding disorder in his maternal uncle. His temperature is 36.9°C (98.4°F). The left knee is erythematous, swollen, and tender; range of motion is limited. No other joints are affected. An x-ray of the knee shows an effusion but no structural abnormalities of the joint. Arthrocentesis is conducted. The synovial fluid is bloody. Further evaluation of this patient is most likely to show which of the following findings?

Answer Choices:

A. Elevated antinuclear antibody levels B. Decreased platelet count C. Prolonged partial thromboplastin time D. Synovial fluid leukocytosis

Correct Answer: C. Prolonged partial thromboplastin time

Question 125

Clinical Scenario:

A 6-year-old girl is brought to the emergency department with difficulty in breathing. Her parents mention that the child has been experiencing an increasing difficulty in breathing over the past few weeks. It is more

prominent when she plays outside in the garden. She has similar episodes about twice a week. She has had a slight difficulty in breathing in the past, but it used to subside once she was rested. During the last month, she has also woken up breathless a couple of times at night, the last episode having occurred last night. A pulmonologist suspects an intermittent obstructive lung disease and orders a pulmonary function test. Her forced expiratory volume is assessed before and after the administration of inhaled albuterol. Her readings are plotted in the graph below. Based on the graph below, which of the following percentage changes in her expiratory volumes would indicate a reversible obstructive pulmonary condition?

Answer Choices:

A. 12% B. 9% C. 50% D. 75%

Correct Answer: A. 12%

Question 126

Clinical Scenario:

A 74-year-old man is brought to the physician by his wife for progressively worsening confusion and forgetfulness. Vital signs are within normal limits. Physical examination shows a flat affect and impaired short-term memory. An MRI of the brain is shown. Further evaluation of this patient is most likely to show which of the following findings?

Answer Choices:

A. Broad-based gait B. Postural instability C. Pill-rolling tremor D. Choreiform movements

Correct Answer: A. Broad-based gait

Question 127

Clinical Scenario:

A 65-year-old man with chronic obstructive lung disease, depression, and type 2 diabetes mellitus comes to the physician with fever, chills, dyspnea, and a productive cough for 5 days. His temperature is 38.8°C (101.8°F) and respirations are 30/min. An x-ray of the chest shows a right lower lobe infiltrate, and sputum culture grows bacteria that are sensitive to fluoroquinolone antibiotics. Pharmacotherapy with oral moxifloxacin is initiated. Three days later, the patient continues to have symptoms despite being compliant with the antibiotic. Serum moxifloxacin levels are undetectable. The lack of response to antibiotic therapy in this patient is most likely due to the concurrent ingestion of which of the following medications?

Answer Choices:

A. Multivitamin B. Glimepiride C. Theophylline D. Prednisone

Correct Answer: A. Multivitamin

Question 128

Clinical Scenario:

A 47-year-old woman with chronic epigastric pain comes to the physician because of a 1-month history of intermittent, loose, foul-smelling stools. She has also had a 6-kg (13-lb) weight loss. She has consumed 9–10 alcoholic beverages daily for the past 25 years. Seven years ago, she traveled to Mexico on vacation; she has not been outside the large metropolitan area in which she resides since then. She appears malnourished. The stool is pale and loose; fecal fat content is elevated. An immunoglobulin A serum anti-tissue transglutaminase antibody assay is negative. Further evaluation is most likely to show which of the following?

Answer Choices:

A. Trophozoites on stool microscopy B. Pancreatic calcifications C. Villous atrophy of duodenal mucosa D. Positive lactulose breath test

Correct Answer: B. Pancreatic calcifications

Question 129

Clinical Scenario:

A 27-year-old woman with sickle cell disease and at 39-weeks' gestation is brought to the emergency department in active labor. She has had multiple episodes of acute chest syndrome and has required several transfusions in the past. She has a prolonged vaginal delivery complicated by postpartum bleeding, and she receives a transfusion of 1 unit of packed red blood cells. One hour later, the patient experiences acute flank pain. Her temperature is 38.7°C (101.6°F), pulse is 115/min, respirations are 24/min, and blood pressure is 95/55 mm Hg. Foley catheter shows dark brown urine. Further evaluation of this patient is most likely to show which of the following?

Answer Choices:

A. Serum antibodies against class I HLA antigens B. Positive direct Coombs test C. Positive blood cultures D. Low levels of serum IgA immunoglobulins

Correct Answer: B. Positive direct Coombs test

Question 130

Clinical Scenario:

A 26-year-old woman comes to the physician because of a progressive swelling in her mouth that she first noticed 5 years ago. Initially, the swelling was asymptomatic but has now caused some difficulty while chewing food for the past month. She has no pain. She has not undergone any dental procedures in the past 5 years. She has bronchial asthma. Her only medication is an albuterol inhaler. She appears healthy. Her temperature is 37°C (98.6°F), pulse is 70/min, and blood pressure is 110/70 mm Hg. Examination shows a 1.5-cm smooth, unilobular, bony hard, nontender mass in the midline of the hard palate. There is no cervical or

submandibular lymphadenopathy. The remainder of the examination shows no abnormalities. Which of the following is the most likely diagnosis?

Answer Choices:

A. Palatal pleomorphic adenoma B. Necrotizing sialometaplasia C. Nasopalatine duct cyst D. Torus palatinus "

Correct Answer: D. Torus palatinus "

Question 131

Clinical Scenario:

A 58-year-old man comes to the physician because of a 5-day history of progressively worsening shortness of breath and fatigue. He has smoked 1 pack of cigarettes daily for 30 years. His pulse is 96/min, respirations are 26/min, and blood pressure is 100/60 mm Hg. An x-ray of the chest is shown. Which of the following is the most likely cause of this patient's findings?

Answer Choices:

A. Left ventricular failure B. Tricuspid regurgitation C. Pulmonary embolism D. Interstitial pneumonia

Correct Answer: A. Left ventricular failure

Question 132

Clinical Scenario:

A 52-year-old G3P3 presents to her gynecologist complaining of painful intercourse. She reports a 6-month history of intermittent dryness, itching, and burning during intercourse. Upon further questioning, she also reports poor sleep and occasional periods during which she feels very warm and sweats profusely. Her past medical history is significant for poorly controlled hypertension and a 10 pack-year smoking history. She takes hydrochlorothiazide and enalapril. Her temperature is 99.3°F (37.4°C), blood pressure is 135/85 mmHg, pulse is 90/min, and respirations are 18/min. On examination, she is a healthy female in no distress. Pelvic examination reveals no adnexal or cervical motion tenderness. Which of the following sets of hormone levels are most likely to be found in this patient?

Answer Choices:

A. Increased estrogen, decreased FSH, decreased LH, decreased GnRH B. Decreased estrogen, decreased FSH, decreased LH, increased GnRH C. Decreased estrogen, increased FSH, increased LH, increased GnRH D. Normal estrogen, normal FSH, normal LH, normal GnRH

Correct Answer: C. Decreased estrogen, increased FSH, increased LH, increased GnRH

Question 133

Clinical Scenario:

A 28-year-old woman, gravida 1, para 0, at 10 weeks gestation comes to the physician for her first prenatal visit. Today, she feels well. She has no history of serious illness. Her pulse is 75/min and blood pressure is 110/74 mm Hg. Examination shows no abnormalities. Ultrasonography shows a pregnancy consistent in size with a 10-week gestation. Serum studies in this patient are most likely to show which of the following sets of laboratory values? \$\$\$ Thyroid-binding globulin %%% Free Triiodothyronine (T3) %%% Free Thyroxine (T4) %%% Total T3+T4 \$\$\$

Answer Choices:

A. ↑ normal normal ↑ B. ↓ normal normal ↓ C. Normal normal normal normal D. ↓ ↓ normal ↓

Correct Answer: A. ↑ normal normal ↑

Question 134

Clinical Scenario:

A 55-year-old man presents to his primary care physician for a wellness checkup. He states that he generally feels well and has no complaints at this time. The patient consumes alcohol frequently, eats a high sodium diet, and is sedentary. His temperature is 97.5°F (36.4°C), blood pressure is 167/108 mmHg, pulse is 90/min, respirations are 15/min, and oxygen saturation is 99% on room air. The patient's blood pressure at his last 2 appointments were 159/100 mmHg and 162/99 mmHg, respectively. His physician wants to intervene to manage his blood pressure. Which of the following is the most effective treatment for this patient's hypertension?

Answer Choices:

A. DASH diet B. Reduce alcohol consumption C. Sodium restriction D. Weight loss

Correct Answer: D. Weight loss

Question 135

Clinical Scenario:

A 45-year-old woman comes to see you for a second opinion regarding an upcoming surgery for pancreatic insulinoma. While taking a surgical history, she tells you she previously had a pituitary tumor resected. For which additional neoplasms might you consider testing her?

Answer Choices:

A. Medullary thyroid carcinoma B. Pheochromocytoma C. Parathyroid adenoma D. Multiple myeloma

Correct Answer: C. Parathyroid adenoma

Question 136

Clinical Scenario:

A 34-year-old G3P2 presents at 33 weeks gestation with vaginal bleeding that started last night while she was asleep. She denies uterine contractions or abdominal pain. She had a cesarean delivery in her previous pregnancy. She also reports a 10 pack-year smoking history. The vital signs are as follows: blood pressure, 130/80 mm Hg; heart rate, 84/min; respiratory rate, 12/min; and temperature, 36.8°C (98.2°F). The physical examination is negative for abdominal tenderness or palpable uterine contractions. The perineum is mildly bloody. On speculum examination, no vaginal or cervical lesions are seen. A small amount of blood continues to pass through the cervix. Which of the following findings would you expect on ultrasound examination?

Answer Choices:

A. Partial covering of the internal cervical os by the placental edge B. Retroplacental blood accumulation C. Placental calcification D. Cysts on the placental surface

Correct Answer: A. Partial covering of the internal cervical os by the placental edge

Question 137

Clinical Scenario:

A previously healthy 46-year-old woman comes to her physician because of an itchy rash on her legs. She denies any recent trauma, insect bites, or travel. Her vital signs are within normal limits. Examination of the oral cavity shows white lace-like lines on the buccal mucosa. A photograph of the rash is shown. A biopsy specimen of the skin lesion is most likely to show which of the following?

Answer Choices:

A. Decreased thickness of the stratum granulosum B. Lymphocytes at the dermoepidermal junction C. Proliferation of vascular endothelium D. Deposition of antibodies around epidermal cells

Correct Answer: B. Lymphocytes at the dermoepidermal junction

Question 138

Clinical Scenario:

A 36-year-old primigravid woman at 15 weeks' gestation comes to the physician for a routine prenatal visit. She has not been taking prenatal vitamins and admits to consuming alcohol regularly. Pelvic examination shows a uterus consistent in size with a 15-week gestation. A quadruple screening test shows markedly elevated maternal serum α -fetoprotein. Maternal serum concentrations of β -human chorionic gonadotropin, estriol, and inhibin A are normal. Which of the following is the most likely explanation for these findings?

Answer Choices:

A. Trisomy 21 B. Holoprosencephaly C. Spina bifida cystica D. Trisomy 18

Correct Answer: C. Spina bifida cystica

Question 139

Clinical Scenario:

Six days after being admitted to the hospital for a cholecystectomy, a 56-year-old woman has high-grade fevers, chills, malaise, and generalized weakness. She has been hospitalized twice in the last year for acute cholecystitis. She had a molar extraction around 2 weeks ago. Her last colonoscopy was 8 months ago and showed a benign polyp that was removed. She has mitral valve prolapse, hypertension, rheumatoid arthritis, and hypothyroidism. Current medications include metformin, rituximab, levothyroxine, and enalapril. Her temperature is 38.3°C (101°F), pulse is 112/min, and blood pressure is 138/90 mm Hg. Examination shows painless macules over her palms and soles and linear hemorrhages under her nail beds. The lungs are clear to auscultation. There is a grade 3/6 systolic murmur heard best at the apex. Blood is drawn and she is started on intravenous antibiotic therapy. Two sets of blood cultures grow coagulase-negative staphylococci. An echocardiography shows a large oscillating vegetation on the mitral valve and moderate mitral regurgitation. Which of the following is the strongest predisposing factor for this patient's condition?

Answer Choices:

A. Predamaged heart valve B. Recent dental procedure C. Immunosuppression D. Infected peripheral venous catheter

Correct Answer: D. Infected peripheral venous catheter

Question 140

Clinical Scenario:

A 9-year-old boy presents with abdominal pain that started nearly 6 hours ago. The pain is located in the periumbilical area and radiates to the right lower quadrant. There was no vomiting or passage of stool since the onset of pain. The patient reports that he passed stools with blood several times during the past month. The vital signs include: blood pressure 110/70 mm Hg, heart rate 81/min, respiratory rate 16/min, and temperature 37.5°C (99.5°F). The physical examination reveals abdominal tenderness and guarding in the periumbilical area. An abdominal ultrasound does not reveal an appendiceal abnormality. The patient underwent an exploratory laparoscopy. At the time of laparoscopy, a 2-cm wide inflamed diverticulum is found 40 cm proximal to the ileocecal valve. The diverticulum is resected and sent for histologic evaluation. The result is shown in the exhibit. Which of the following statements is true?

Answer Choices:

A. It resulted from traction by embryonic peritoneal adhesions. B. The most probable cause of this diverticulum is increased intraluminal pressure. C. It is an unobliterated embryonic bile duct. D. It is a remnant of the embryonic omphalomesenteric duct.

Correct Answer: D. It is a remnant of the embryonic omphalomesenteric duct.

Question 141

Clinical Scenario:

A 48-year-old man comes to the physician because of a 3-month history of worsening shortness of breath and cough productive of frothy, whitish sputum. One year ago, he had a similar episode lasting 6 months. He has smoked a pack of cigarettes daily for 25 years. Physical examination shows bluish discoloration of the tongue and lips. Scattered expiratory wheezing and rhonchi are heard throughout both lung fields. Further evaluation of this patient is most likely to show which of the following findings?

Answer Choices:

A. Increased pulmonary capillary wedge pressure B. Normal FEV1 C. Increased FEV1/FVC ratio D. Increased serum hematocrit

Correct Answer: D. Increased serum hematocrit

Question 142

Clinical Scenario:

A 71-year-old African American man is brought to the emergency department with sudden onset lower limb paralysis and back pain. He has had generalized bone pain for 2 months. He has no history of severe illnesses. He takes ibuprofen for pain. On examination, he is pale. The vital signs include: temperature 37.1°C (98.8°F), pulse 68/min, respiratory rate 16/min, and blood pressure 155/90 mm Hg. The neurologic examination shows paraparesis. The 8th thoracic vertebra is tender to palpation. X-ray of the thoracic vertebrae confirms a compression fracture at the same level. The laboratory studies show the following: Laboratory test Hemoglobin 9 g/dL Mean corpuscular volume 95 μm^3 Leukocyte count 5,000/mm³ Platelet count 240,000/mm³ ESR 85 mm/hr Serum Na⁺ 135 mEq/L K⁺ 4.2 mEq/L Cl⁻ 113 mEq/L HCO₃⁻ 20 mEq/L Ca⁺ 11.8 mg/dL Albumin 4 g/dL Urea nitrogen 38 mg/dL Creatinine 2.2 mg/dL Which of the following is the most likely mechanism underlying this patient's vertebral fracture?

Answer Choices:

A. Acidosis-induced bone lysis B. Increased mechanical pressure C. Increased osteoblastic activity D. Proliferation of tumor cells

Correct Answer: D. Proliferation of tumor cells

Question 143

Clinical Scenario:

A 67-year-old male presents to his primary care physician complaining of increased fatigue over the last year. He also says that his friends say he appears to be more pale. His past medical history is significant for 10 years of arthritis. Physical exam reveals spoon shaped nails as well as conjunctival pallor. Based on clinical suspicion RBC tests are ordered showing an mean corpuscular volume (MCV) of 75 fl (normal 80-100 fl) and a

peripheral blood smear is obtained and found to be normal. Iron studies shows a serum iron of 30 micromolar (normal range 50-170) and a serum ferritin of 300 micrograms/liter (normal range 15-200). What is the most likely diagnosis in this patient?

Answer Choices:

A. Anemia of chronic disease B. Beta-thalassemia C. Iron deficiency anemia D. Lead poisoning anemia

Correct Answer: A. Anemia of chronic disease

Question 144

Clinical Scenario:

A 45-year-old man is brought to the emergency department by his friends because of a 1-hour history of shortness of breath and squeezing chest pain. They were at a party where cocaine was consumed. A diagnosis of acute myocardial infarction is made. The physician stabilizes the patient and transfers him to the inpatient unit. Six hours later, his wife arrives at the emergency department and requests information about her husband's condition. Which of the following is the most appropriate action by the physician?

Answer Choices:

A. Obtain authorization from the patient to release information B. Inform the wife about her husband's condition C. Consult the hospital ethics committee D. Request the patient's durable power of attorney document

Correct Answer: A. Obtain authorization from the patient to release information

Question 145

Clinical Scenario:

A 36-year-old woman comes to the physician because of new onset limping. For the past 2 weeks, she has had a tendency to trip over her left foot unless she lifts her left leg higher while walking. She has not had any trauma to the leg. She works as a flight attendant and wears compression stockings to work. Her vital signs are within normal limits. Physical examination shows weakness of left foot dorsiflexion against minimal resistance. There is reduced sensation to light touch over the dorsum of the left foot, including the web space between the 1st and 2nd digit. Further evaluation is most likely to show which of the following?

Answer Choices:

A. Decreased ankle jerk reflex B. Normal foot eversion C. Normal foot inversion D. Weak hip flexion

Correct Answer: C. Normal foot inversion

Question 146

Clinical Scenario:

Researchers are investigating the mechanism of cell apoptosis and host defense in mice. They have observed that mice with certain gene deletions are not able to fight the induced viral infection. They identify a cell that is able to destroy target cells infected with viruses by exocytosis of granule contents, which induces the activation of caspases. Which type of cell is responsible for this process?

Answer Choices:

A. Macrophages B. Neutrophils C. CD8+ lymphocytes D. Eosinophils

Correct Answer: C. CD8+ lymphocytes

Question 147

Clinical Scenario:

A 40-year-old man is rushed to the emergency department after being involved in a motor vehicle accident. He has lacerations on his right arm and some minor abrasions on his face and lower limbs. The resident on call quickly manages the patient with proper care of his open wounds in the emergency department. The patient is admitted to the surgery unit for the daily care of his wounds. His lacerations begin to heal with proper dressing and occasional debridement. Which of the following best describes the healing process in this patient?

Answer Choices:

A. Abundant lymphocytes accumulate during the healing process, forming a granuloma. B. Healing by secondary intention will occur along with the deposition of large amount of granulation tissue. C. Healing involves abscess formation, which should be drained. D. Healing by primary intention will occur without granulation tissue formation.

Correct Answer: B. Healing by secondary intention will occur along with the deposition of large amount of granulation tissue.

Question 148

Clinical Scenario:

A 15-year-old girl comes to the physician with her father for evaluation of her tall stature. She is concerned because she is taller than all of her friends. Her birth weight and height were within normal limits. Her father is 174 cm (5 ft 7 in) tall; her mother is 162 cm (5 ft 3 in) tall. She is at the 98th percentile for height and 90th percentile for BMI. She has not had her menstrual period yet. Her mother has Graves disease. Vital signs are within normal limits. Examination shows a tall stature with broad hands and feet. There is frontal bossing and protrusion of the mandible. Finger perimetry is normal. The remainder of the examinations shows no abnormalities. Serum studies show a fasting serum glucose of 144 mg/dL. An x-ray of the left hand and wrist shows a bone age of 15 years. Which of the following is most likely to have prevented this patient's condition?

Answer Choices:

A. Transsphenoidal adenomectomy B. Letrozole therapy C. Methimazole therapy D. Caloric restriction

Correct Answer: A. Transsphenoidal adenomectomy

Question 149

Clinical Scenario:

A 45-year-old gravida 1, para 0 woman is brought to the hospital in labor at 39 weeks gestation. After 4 hours, she gives birth to a healthy appearing baby boy with APGAR scores of 7 at 1 minute and 9 at 5 minutes. She had limited prenatal screening but did have an ultrasound at 35 weeks that showed polyhydramnios. The next day, the neonate vomits greenish-yellow fluid after breastfeeding. This occurs 2 more times that day after feeding and several times between feedings. The next day, the neonate appears weak with difficulty latching to the breast and is dehydrated. The physician on duty is concerned and orders blood work, IV fluids, and the abdominal X-ray shown below. Which of the following disorders is most associated with the newborn's condition?

Answer Choices:

A. Trisomy 13 B. Trisomy 21 C. 47 XXY D. 45 XO

Correct Answer: B. Trisomy 21

Question 150

Clinical Scenario:

A 38-year-old man comes to the physician because of severe muscle pain and swelling of his eyelids for 3 days. He has also had fever and chills during this period. For the last 2 days, he has had severe pain in his mouth while chewing. He had an episode of diarrhea a month ago for which he did not seek medical care. He has no history of serious illness. His sister has dermatomyositis. He returned from a hunting trip to eastern Europe 45 days ago. His temperature is 38.1°C (100.6°F), pulse is 80/min, and blood pressure is 130/70 mm Hg. Examination shows periorbital edema and severe generalized muscle tenderness. There are splinter hemorrhages on both hands. Laboratory studies show: Hemoglobin 14.2 g/dL Leukocyte count 12,500/mm³ Segmented neutrophils 60% Eosinophils 18% Lymphocytes 20% Monocytes 2% Serum Glucose 117 mg/dL Creatinine 1.1 mg/dL Alkaline phosphatase 72 U/L Creatine kinase 765 U/L Urinalysis is within normal limits. Which of the following is most likely to have prevented this patient's condition?"

Answer Choices:

A. Clean drinking water B. Cooking meat to 71°C (160°F) C. Consume pasteurized dairy products D. Metronidazole at the onset of diarrhea

Correct Answer: B. Cooking meat to 71°C (160°F)

Question 151

Clinical Scenario:

A 32-year-old woman who recently emigrated to the USA from Japan comes to the physician because of a 3-month history of night sweats, malaise, and joint pain. During this time, she has also had a 6-kg (13-lb) weight loss. Physical examination shows weak brachial and radial pulses. There are tender subcutaneous nodules on both legs. Carotid bruits are heard on auscultation bilaterally. Laboratory studies show an erythrocyte sedimentation rate of 96 mm/h. A CT scan of the chest shows thickening and narrowing of the aortic arch. Microscopic examination of the aortic arch is most likely to show which of the following findings?

Answer Choices:

A. Fibrinoid necrosis of the intima and media B. Granulomatous inflammation of the media C. Subendothelial immune complex deposition D. Calcification of the media

Correct Answer: B. Granulomatous inflammation of the media

Question 152

Clinical Scenario:

A genetic population study is being conducted to find the penetrance of a certain disease. This disease is associated with impaired iron metabolism and primarily affects the liver. Patients often present with diabetes and bronze skin pigmentation. After a genetic screening of 120 inhabitants with a family history of this disease, 40 were found to have the disease-producing genotype, but only 10 presented with symptoms. What are the chances of the screened patients with said genotype developing the disease phenotype?

Answer Choices:

A. 3% B. 25% C. 40% D. 0.4%

Correct Answer: B. 25%

Question 153

Clinical Scenario:

A 28-year-old female patient with a history of schizophrenia, type 2 diabetes mellitus, and hypothyroidism comes to clinic stating she would like to be put back on a medication. She recently stopped taking her haloperidol as it made it hard for her to "sit still." She requests to be put on olanzapine as a friend from a support group said it was helpful. Why should this medication be avoided in this patient?

Answer Choices:

A. There is a high risk for retinopathy B. The patient has type 2 diabetes C. The patient may develop galactorrhea D. Tardive dyskinesia will likely result from the prolonged use of olanzapine

Correct Answer: B. The patient has type 2 diabetes

Question 154

Clinical Scenario:

A 65-year-old man comes to the physician for a routine health maintenance examination. He has a strong family history of colon cancer. A screening colonoscopy shows a 4 mm polyp in the upper sigmoid colon. Which of the following findings on biopsy is associated with the lowest potential for malignant transformation into colorectal carcinoma?

Answer Choices:

A. Tree-like branching of muscularis mucosa B. Regenerating epithelium with inflammatory infiltrate C. Finger-like projections with a fibrovascular core D. Hyperplastic epithelium at the base of crypts

Correct Answer: D. Hyperplastic epithelium at the base of crypts

Question 155

Clinical Scenario:

A 38-year-old man comes to the physician because of a 2-week history of abdominal pain and an itchy rash on his buttocks. He also has fever, nausea, and diarrhea with mucoid stools. One week ago, the patient returned from Indonesia, where he went for vacation. Physical examination shows erythematous, serpiginous lesions located in the perianal region and the posterior thighs. His leukocyte count is 9,000/mm³ with 25% eosinophils. Further evaluation is most likely to show which of the following findings?

Answer Choices:

A. Rhabditiform larvae on stool microscopy B. Oocysts on acid-fast stool stain C. Giardia lamblia antibodies on stool immunoassay D. Branching septate hyphae on KOH preparation

Correct Answer: A. Rhabditiform larvae on stool microscopy

Question 156

Clinical Scenario:

A 22-year-old man seeks help from a physician for his heroin addiction. He tells the doctor that he started using heroin at the age of 17 and gradually started increasing the dose. He has been trying to quit for the last 6 months after realizing the negative consequences of his addiction but has not succeeded because of the withdrawal symptoms. The physician suggests a drug that can be taken within a supervised rehabilitation program as a substitute for heroin to help alleviate withdrawal symptoms. The drug will then be tapered over time. He is further informed by the physician that this drug is not to be taken by the patient on his own and will not work in an emergency situation related to heroin withdrawal. Which of the following drugs is most likely to have been recommended by the physician?

Answer Choices:

A. Codeine B. Clonidine C. Methadone D. Naloxone

Correct Answer: C. Methadone

Question 157

Clinical Scenario:

An 18-year-old high school student is brought to the emergency department after collapsing during her cheerleading practice session. She was brought to the hospital even though she regained consciousness within seconds. Her mother informs the doctor that she had a similar episode last month at a party, but they thought it was due to stress and exhaustion. Both incidents occurred in a loud and crowded environment. Her past medical history is insignificant. Her blood pressure is 120/80 mm Hg and the pulse is 77/min and regular. Physical examination findings are within normal limits. A set of tests are ordered along with a 12-lead ECG. The ECG tracing obtained is shown. What is the best treatment option for this patient?

Answer Choices:

A. Erythromycin B. Furosemide C. Nadolol D. Quinidine

Correct Answer: C. Nadolol

Question 158

Clinical Scenario:

A 42-year-old man is referred for an endocrinology consult because of decreased triiodothyronine (T3) hormone levels. He presented to the emergency department 1 week prior to this consultation with pneumonia and was admitted to a medicine service for management of his infection. He has since recovered from his infection after intravenous antibiotic administration. He currently has no symptoms and denies feeling cold or lethargic. A panel of laboratory tests are obtained with the following results:

Thyroid-stimulating hormone: 4.7 μ U/mL Thyroxine (T4): 6 μ g/dL Triiodothyronine (T3): 68 ng/dL

Which of the following additional findings would most likely also be seen in this patient?

Answer Choices:

A. Decreased free T3 concentration B. Increased free T3 concentration C. Increased reverse T3 concentration D. Normal free and reverse T3 concentration

Correct Answer: C. Increased reverse T3 concentration

Question 159

Clinical Scenario:

A 55-year-old woman comes to the physician because of a 6-month history of worsening shortness of breath on exertion and fatigue. She has type 1 diabetes mellitus, hypertension, hypercholesterolemia, and chronic kidney disease. Her mother was diagnosed with colon cancer at the age of 65 years. Her blood pressure is 145/92 mm

Hg. Examination shows conjunctival pallor. Laboratory studies show: Hemoglobin 9.2 g/dL Mean corpuscular volume 88 μm^3 Reticulocyte count 0.6 % Serum Ferritin 145 ng/mL Creatinine 3.1 mg/dL Calcium 8.8 mg/dL A fecal occult blood test is pending. Which of the following is the most likely underlying cause of this patient's symptoms?"

Answer Choices:

A. Autoantibodies against the thyroid gland B. Chronic occult blood loss C. Deficient vitamin B12 intake D. Decreased erythropoietin production

Correct Answer: D. Decreased erythropoietin production

Question 160

Clinical Scenario:

A 60-year-old male comes to the emergency department because of a 3-day history of intermittent shortness of breath and palpitations. The episodes are unprovoked and occur randomly. The day before, he felt lightheaded while walking and had to sit down abruptly to keep from passing out. He has hypertension and coronary artery disease. Cardiac catheterization 5 years ago showed occlusion of the left anterior descending artery, and he underwent placement of a stent. Current medications include aspirin, metoprolol, lisinopril, and clopidogrel. He does not drink alcohol or use any illicit drugs. He has smoked one-half pack of cigarettes daily for 20 years. He appears well. His temperature is 37°C (98.6°F), pulse is 136/min, respirations are 18/min, and blood pressure is 110/85 mm Hg. The lungs are clear to auscultation. Cardiac examination shows a rapid, irregular rhythm. Shortly after, an ECG is performed. Which of the following is the most likely cause of this patient's findings?

Answer Choices:

A. Premature ventricular contractions B. Abnormal automaticity within the ventricle C. Degeneration of sinoatrial node automaticity D. Wandering atrial pacemaker

Correct Answer: C. Degeneration of sinoatrial node automaticity

Question 161

Clinical Scenario:

A 68-year-old woman comes to the physician for evaluation of diminished vision for several months. Twenty-eight years ago, she was diagnosed with systemic lupus erythematosus, which has been well controlled with hydroxychloroquine. Fundoscopic examination shows concentric rings of hypopigmentation and hyperpigmentation surrounding the fovea bilaterally. Visual field examination of this patient is most likely to show which of the following findings?

Answer Choices:

A. Bitemporal hemianopia B. Right monocular blindness C. Paracentral scotoma D. Binasal hemianopia

Correct Answer: C. Paracentral scotoma

Question 162

Clinical Scenario:

A 3900-g (8.6-lb) male infant is delivered at 39 weeks' gestation via spontaneous vaginal delivery. Pregnancy and delivery were uncomplicated but a prenatal ultrasound at 20 weeks showed a defect in the pleuroperitoneal membrane. Further evaluation of this patient is most likely to show which of the following findings?

Answer Choices:

A. Gastric fundus in the thorax B. Pancreatic ring around the duodenum C. Hypertrophy of the gastric pylorus D. Large bowel in the inguinal canal

Correct Answer: A. Gastric fundus in the thorax

Question 163

Clinical Scenario:

A 25-year-old man presents to the emergency department with bilateral eye pain. The patient states it has slowly been worsening over the past 48 hours. He admits to going out this past weekend and drinking large amounts of alcohol and having unprotected sex but cannot recall a predisposing event. The patient's vitals are within normal limits. Physical exam is notable for bilateral painful and red eyes with opacification and ulceration of each cornea. The patient's contact lenses are removed and a slit lamp exam is performed and shows bilateral corneal ulceration. Which of the following is the best treatment for this patient?

Answer Choices:

A. Acyclovir B. Gatifloxacin eye drops C. Intravitreal vancomycin and ceftazidime D. Topical dexamethasone and refrain from wearing contacts

Correct Answer: B. Gatifloxacin eye drops

Question 164

Clinical Scenario:

A 26-year-old man from India visits the clinic with complaints of feeling tired all the time and experiencing lack of energy for the past couple of weeks. He also complains of weakness and numbness of his lower limbs. He has been strictly vegan since the age of 18, including not consuming eggs and milk. He does not take any vitamin or dietary supplements. Physical examination reveals a smooth, red beefy tongue along with lower extremity sensory and motor deficits. What other finding is most likely to accompany this patient's condition?

Answer Choices:

A. Ataxia B. Psychiatric symptoms C. Decreased visual acuity D. Microcytic anemia

Correct Answer: A. Ataxia

Question 165

Clinical Scenario:

A 68-year-old man with alcohol use disorder is brought to the physician by his sister for frequent falls and an unsteady gait over the past 2 months. He has not seen a physician in 10 years. He appears emaciated and inattentive. He is oriented to person only. Physical examination shows a wide-based gait with slow, short steps. Eye examination shows lateral gaze paralysis and horizontal nystagmus. One month later, he dies. Which of the following is the most likely finding on autopsy?

Answer Choices:

A. Small vessel hemorrhage in mammillary bodies B. Depigmentation of the substantia nigra C. Widespread atrophy of cerebral cortex D. Atrophy of the caudate and putamen

Correct Answer: A. Small vessel hemorrhage in mammillary bodies

Question 166

Clinical Scenario:

A 68-year-old woman comes to the emergency department because of abdominal pain for 3 days. Physical examination shows guarding and tenderness to palpation over the left lower abdomen. Test of the stool for occult blood is positive. A CT scan of the abdomen is shown. Which of the following mechanisms best explains the patient's imaging findings?

Answer Choices:

A. Failed neural crest cell migration B. Abnormal organ rotation C. Abnormal outpuching of hollow organ D. Impaired organ ascent

Correct Answer: D. Impaired organ ascent

Question 167

Clinical Scenario:

A 33-year-old man is brought to the emergency department after being involved in a bar fight. Physical examination shows tenderness to palpation over the left side of the back. An x-ray of the chest shows a fracture of the 12th rib on the left side. Further evaluation is most likely to show which of the following injuries?

Answer Choices:

A. Liver hematoma B. Colon perforation C. Kidney laceration D. Pancreatic transection

Correct Answer: C. Kidney laceration

Question 168

Clinical Scenario:

A 72-year-old man is brought to the emergency department because of progressive weakness of his lower extremities and urinary incontinence for the past 3 weeks. Over the past 2 months, he has also had increasing back pain. Physical examination shows an unsteady gait. Muscle strength is decreased in both lower extremities. Sensation to pain, temperature, and position sense is absent in the buttocks, perineum, and lower extremities. Ankle clonus is present. An x-ray of the spine shows multiple sclerotic lesions in the thoracic and lumbar vertebrae. Further evaluation of this patient is most likely to show which of the following?

Answer Choices:

A. Elevated prostate-specific antigen in the serum B. Palpable thyroid nodule on neck examination C. Elevated carcinoembryonic antigen in the serum D. Irregular, asymmetric mole on skin examination "

Correct Answer: A. Elevated prostate-specific antigen in the serum

Question 169

Clinical Scenario:

A 17-year-old boy is brought to the physician because of a sore throat, nonproductive cough, and bloody urine for 3 days. He has had 2 similar episodes involving a sore throat and bloody urine over the past year. His sister has systemic lupus erythematosus. His temperature is 38.1°C (100.6°F). Serum studies show a urea nitrogen concentration of 8 mg/dL and a creatinine concentration of 1.4 mg/dL. Urinalysis shows acanthocytes and red blood cell casts. Renal ultrasonography shows no abnormalities. A renal biopsy is most likely to show which of the following findings?

Answer Choices:

A. IgA mesangial deposition B. Capillary wire looping C. Splitting of the glomerular basement membrane D. Effacement of the foot processes

Correct Answer: A. IgA mesangial deposition

Question 170

Clinical Scenario:

A 25-year-old woman, gravida 2, para 1, at 24 weeks' gestation comes to the physician for a prenatal visit. She reports feeling fatigue and having swollen legs lately. One month ago, she had a low-grade fever, a runny nose, painful joints, and a sore throat that resolved spontaneously. Pregnancy and delivery of her first child were uncomplicated. She does not smoke or drink alcohol. She does not use illicit drugs. Medications include folic acid and a multivitamin. Vital signs are within normal limits. Pelvic examination shows a uterus consistent in size with a 24-week gestation. There is bilateral edema around the ankles. Pelvic ultrasonography shows fluid

accumulation within the fetal scalp and signs of pleural effusions bilaterally. Which of the following is the most likely underlying cause of these findings?

Answer Choices:

A. Herpes simplex virus B. Parvovirus B19 C. *Listeria monocytogenes* D. *Toxoplasma gondii* "

Correct Answer: B. Parvovirus B19

Question 171

Clinical Scenario:

A 62-year-old man presents to the emergency room with an acute myocardial infarction. Twenty-four hours after admission to the cardiac intensive care unit, he develops oliguria. Laboratory tests show that his serum BUN is 59 mg/dL and his serum creatinine is 6.2 mg/dL. Renal biopsy reveals necrosis of the proximal tubules and thick ascending limb of Henle's loop. Which of the following would you most likely observe on a microscopic examination of this patient's urine?

Answer Choices:

A. Fatty casts B. Muddy brown casts C. Hyaline casts D. Broad waxy casts

Correct Answer: B. Muddy brown casts

Question 172

Clinical Scenario:

Thirty minutes after vaginal delivery of a 2780-g (6-lb 2-oz) newborn at term, a 25-year-old woman, gravida 1, para 1, has heavy vaginal bleeding. Her pregnancy was complicated by pre-eclampsia. Her pulse is 111/min and blood pressure is 95/65 mm Hg. Physical examination shows a fundal height 2 inches below the xiphoid process of the sternum. A drug with which of the following mechanisms of action is most appropriate for this patient?

Answer Choices:

A. Inhibition of norepinephrine reuptake B. Activation of phospholipase C C. Increased synthesis of cyclic AMP D. Binding to prostaglandin I₂ receptors

Correct Answer: B. Activation of phospholipase C

Question 173

Clinical Scenario:

A 25-year-old G1P0 at 20 weeks of gestation woman arrives at a prenatal appointment complaining of pelvic pressure. She has had an uncomplicated pregnancy thus far. She takes prenatal vitamins and eats a well-

balanced diet. Her medical history is significant for major depressive disorder that has been well-controlled on citalopram. Her mother had gestational diabetes with each of her 3 pregnancies. On physical exam, the cervix is soft and closed with minimal effacement. There is white vaginal discharge within the vagina and vaginal vault without malodor. Vaginal pH is 4.3. A transvaginal ultrasound measures the length of the cervix as 20 mm. Which of the following is most likely to prevent preterm birth in this patient?

Answer Choices:

A. Metformin B. Metronidazole C. Prednisone D. Vaginal progesterone

Correct Answer: D. Vaginal progesterone

Question 174

Clinical Scenario:

A 24-year-old female medical student presents to the emergency department after she develops sudden difficulty breathing and vague chest pain while preparing for exams. The chest pain is non-pleuritic without radiation. She denies any recent travel. She denies any hemoptysis, nausea, vomiting, or leg pain. She only takes oral contraceptives; she denies smoking or alcohol use. Her vitals reveal a heart rate of 120 beats per minute, blood pressure of 100/80 mm Hg, and respiratory rate of 30 per minute. She is afebrile. Otherwise, her physical exam is unremarkable. A CT scan of her chest with IV contrast reveals filling defects along her left pulmonary artery. Which of the following is the most likely mechanism of this finding?

Answer Choices:

A. Venous stasis B. Endothelial injury C. Hypercoagulability D. Anxiety

Correct Answer: C. Hypercoagulability

Question 175

Clinical Scenario:

A 31-year-old man presents to the Emergency Department with severe left leg pain and paresthesias 4 hours after his leg got trapped by the closing door of a bus. Initially, he had a mild pain which gradually increased to unbearable levels. Past medical history is noncontributory. In the Emergency Department, his blood pressure is 130/80 mm Hg, heart rate is 87/min, respiratory rate is 14/min, and temperature is 36.8°C (98.2°F). On physical exam, his left calf is firm and severely tender on palpation. The patient cannot actively dorsiflex his left foot, and passive dorsiflexion is limited. Posterior tibial and dorsalis pedis pulses are 2+ in the right leg and 1+ in the left leg. Axial load does not increase the pain. Which of the following is the best next step in the management of this patient?

Answer Choices:

A. Lower limb X-ray in two projections B. Lower limb ultrasound C. Splinting and limb rest D. Fasciotomy

Correct Answer: D. Fasciotomy

Question 176

Clinical Scenario:

A 63-year-old woman is brought to the emergency department 1 hour after the onset of right-sided weakness. She was eating breakfast when suddenly she could not lift her spoon. She cried out to her husband but her speech was slurred. For the past 4 months, she has been more anxious than usual and felt fatigued. She used to exercise regularly but had to give up her exercise routine 3 months ago because of lightheadedness and shortness of breath with exertion. She has a history of hypertension. She is a tax accountant and has had increased stress at work recently. She takes lisinopril daily and alprazolam as needed. Her temperature is 37.2°C (99.0°F), pulse is 138/min, respirations are 14/min, and blood pressure is 146/86 mm Hg. Her lungs are clear to auscultation bilaterally and she has an S1 with variable intensity. On neurologic examination, she has a right facial droop and 2/5 strength in the right shoulder, elbow, wrist, and fingers. Sensation is diminished in the right face and arm. Further evaluation is most likely to show which of the following?

Answer Choices:

A. Irregularly irregular rhythm without P waves on ECG B. Left-sided carotid stenosis on duplex ultrasound C. Intraparenchymal hyperdensity on head CT D. Spikes and sharp waves in temporal region on EEG "

Correct Answer: A. Irregularly irregular rhythm without P waves on ECG

Question 177

Clinical Scenario:

A 68-year-old man with type 2 diabetes mellitus comes to the physician because of a 5-month history of episodic palpitations, dizziness, and fatigue. His pulse is 134/min and irregularly irregular, and his blood pressure is 165/92 mm Hg. An ECG shows a narrow complex tachycardia with absent P waves. He is prescribed a drug that decreases the long-term risk of thromboembolic complications by inhibiting the extrinsic pathway of the coagulation cascade. The expected beneficial effect of this drug is most likely due to which of the following actions?

Answer Choices:

A. Inhibit the absorption of vitamin K B. Activate gamma-glutamyl carboxylase C. Activate factor VII calcium-binding sites D. Inhibit the reduction of vitamin K

Correct Answer: D. Inhibit the reduction of vitamin K

Question 178

Clinical Scenario:

A 27-year-old woman presents with painful urination and malodorous urethral discharge. She states she has a single sexual partner and uses condoms for contraception. The patient's blood pressure is 115/80 mm Hg, the heart rate is 73/min, the respiratory rate is 14/min, and the temperature is 36.6°C (97.9°F). Physical

examination shows swelling and redness of the external urethral ostium. There is a yellowish, purulent discharge with an unpleasant odor. The swab culture grows *N. gonorrhoeae*. The doctor explains the diagnosis to the patient, and they discuss the importance of notifying her partner. The patient says she doesn't want her partner to know about her diagnosis and begs the doctor to not inform the health department. She is anxious that everybody will find out that she is infected and that her partner will leave her. She promises they will use barrier contraception while she is treated. Which of the following is the most appropriate course of action?

Answer Choices:

A. Let the patient do as she suggests, because it is her right not to disclose her diagnosis to anyone. B. Explain to the patient that gonorrhea is a mandatory reported disease. C. Refer to the medical ethics committee for consultation. D. Tell the patient that she is required to tell her partner and stress the consequences of untreated gonorrhea in her partner.

Correct Answer: B. Explain to the patient that gonorrhea is a mandatory reported disease.

Question 179

Clinical Scenario:

A 70-year-old woman comes to the physician because of a 4-month history of fatigue, worsening swelling of her ankles, and a 5-kg (11-lb) weight gain. Neurologic examination shows diminished two-point discrimination in her fingers. Laboratory studies show a hemoglobin A1c concentration of 9.2% and a creatinine concentration of 1.3 mg/dL. Urine dipstick shows heavy proteinuria. A biopsy specimen of this patient's kidney is most likely to show which of the following?

Answer Choices:

A. Immune complex deposition B. Interstitial inflammation C. Wire looping of capillaries D. Nodular glomerulosclerosis

Correct Answer: D. Nodular glomerulosclerosis

Question 180

Clinical Scenario:

A 5-year-old boy is brought to the physician because of a painful, burning rash on his left arm for 3 days. Three years ago, he was diagnosed with heart failure due to congenital heart disease and received an allogeneic heart transplantation. He takes cyclosporine to prevent chronic transplant rejection. He has not received any routine childhood vaccinations. A photograph of the rash is shown. Microscopic examination of a skin biopsy specimen is most likely to show which of the following findings?

Answer Choices:

A. Eosinophilic spongiosis and subepidermal blister formation B. Multinucleated epidermal giant cells and intranuclear inclusions C. Papillary microabscesses and granular deposits of IgA D. Fungal hyphae and hyperkeratosis

Correct Answer: B. Multinucleated epidermal giant cells and intranuclear inclusions

Question 181

Clinical Scenario:

A 25-year-old woman comes to the physician because of irregular menstrual bleeding. Menarche occurred at the age of 12 years and menses have occurred at 45 to 90-day intervals. Her last menstrual period was 8 weeks ago. She is not sexually active. Serum studies show: Fasting glucose 178 mg/dL Fasting insulin 29 mcIU/mL (N = 2.6–24.9 mcIU/mL) Luteinizing hormone 160 mIU/mL Total testosterone 3.2 ng/dL (N = 0.06–1.06 ng/dL) Serum electrolytes are within the reference range. Further evaluation of this patient is most likely to show which of the following findings?"

Answer Choices:

A. Elevated serum beta-HCG level B. Adrenal tumor on abdominal MRI C. Enlarged ovaries on transvaginal ultrasound D. Intrastellar mass on cranial contrast MRI

Correct Answer: C. Enlarged ovaries on transvaginal ultrasound

Question 182

Clinical Scenario:

A 10-month-old boy is being treated for a rare kind of anemia and is currently being evaluated for a bone marrow transplant. The patient's mother presents to an appointment with their pediatrician after having done some online research. She has learned that the majority of patients inherit this condition as an autosomal dominant mutation. As a result of the genetic mutation, there is impaired erythropoiesis, leading to macrocytic red blood cells without hypersegmented neutrophils. She also read that children who survive will eventually present with short stature and craniofacial abnormalities. Which of the following is true about this patient's condition?

Answer Choices:

A. Splenectomy is a treatment option B. Occurs due to an inability to convert orotic acid to uridine monophosphate (UMP) C. Fetal hemoglobin level is elevated D. Occurs due to a defect in lymphoblasts and erythroid progenitor cells

Correct Answer: C. Fetal hemoglobin level is elevated

Question 183

Clinical Scenario:

A student health coordinator plans on leading a campus-wide HIV screening program that will be free for the entire undergraduate student body. The goal is to capture as many correct HIV diagnoses as possible with the

fewest false positives. The coordinator consults with the hospital to see which tests are available to use for this program. Test A has a sensitivity of 0.92 and a specificity of 0.99. Test B has a sensitivity of 0.95 and a specificity of 0.96. Test C has a sensitivity of 0.98 and a specificity of 0.93. Which of the following testing schemes should the coordinator pursue?

Answer Choices:

A. Test A on the entire student body followed by Test C on those who are positive B. Test B on the entire student body followed by Test A on those who are positive C. Test C on the entire student body followed by Test A on those who are positive D. Test C on the entire student body followed by Test B on those who are positive

Correct Answer: C. Test C on the entire student body followed by Test A on those who are positive

Question 184

Clinical Scenario:

A 2-year-old boy is brought to the physician because of an increasing productive cough with a moderate amount of white phlegm for the past week. He has been treated for pneumonia with antibiotic therapy four times over the past year. A chest x-ray performed 3 months ago showed no anatomical abnormalities. He has had multiple episodes of bulky greasy stools that don't flush easily. He is at 3rd percentile for height and at 5th percentile for weight. His temperature is 38°C (100.4°F), pulse is 132/min, and respirations are 44/min. A few inspiratory crackles are heard in the thorax. The abdomen is soft and nontender. The remainder of the examination shows no abnormalities. Which of the following is the best initial test to determine the underlying etiology of this patient's illness?

Answer Choices:

A. X-ray of the chest B. Serum immunoglobulin level C. Sweat chloride test D. DNA phenotyping "

Correct Answer: C. Sweat chloride test

Question 185

Clinical Scenario:

A 9-year-old boy is brought to the physician by his parents because of right-sided shoulder pain for 1 day. He has not had chills or sweating. Over the past year, he was treated twice in the emergency department for painful swelling of his hands and feet. He emigrated with his family from Kenya 2 years ago. His temperature is 37.4°C (99.3°F), pulse is 96/min, and blood pressure is 123/82 mm Hg. Physical examination shows no tenderness, erythema, or joint swelling of the shoulder. Laboratory studies show: Hemoglobin 7 g/dL Mean corpuscular volume 88 µm Reticulocyte count 9% Leukocyte count 12,000/mm³ A peripheral blood smear is most likely to show which of the following abnormalities?"

Answer Choices:

A. Teardrop-shaped erythrocytes B. Nuclear remnants in erythrocytes C. Fragmentation of erythrocytes D. Erythrocytes with no central pallor

Correct Answer: B. Nuclear remnants in erythrocytes

Question 186

Clinical Scenario:

A 7-month-old boy is brought to the emergency department by his mother because of a 3-day history of vomiting and poor feeding. The vomit is non-bloody. He transitioned to pureed vegetables 10 days ago. Over the past 2 weeks, he has become increasingly irritable and within the past day has taken more daytime naps and appears much less responsive and interactive. His mother denies any history of fever or trauma at home. He has not received any vaccinations as his parents believe he is already healthy and does not need them. He spends most of the day with a babysitter while both parents are at work. He appears lethargic. His temperature is 37.8°C (100.1°F), pulse is 140/min, respirations are 18/min, and blood pressure is 90/55 mm Hg. The abdomen is soft and nontender. Auscultation of the heart and lungs shows no abnormalities. The anterior fontanelle is tense and bulging. Fundoscopic exam shows bilateral retinal hemorrhage. A complete blood count shows a leukocyte count of 10,000/mm³. An x-ray of the chest shows healing fractures of the 2nd and 3rd right ribs. Further evaluation of this patient is most likely to show which of the following findings?

Answer Choices:

A. Mass in the posterior fossa B. Bacterial infection C. Diffuse axonal damage D. Type I collagen synthesis defect

Correct Answer: C. Diffuse axonal damage

Question 187

Clinical Scenario:

A 70-year-old man presents with severe abdominal pain over the last 24 hours. He describes the pain as severe and associated with diarrhea, nausea, and vomiting. He says he has had a history of postprandial abdominal pain over the last several months. The patient denies any fever, chills, recent antibiotic use. Past medical history is significant for peripheral arterial disease and type 2 diabetes mellitus. The patient reports a 20 pack-year smoking history. His vital signs include blood pressure 90/60 mm Hg, pulse 100/min, respiratory 22/min, temperature 38.0°C (100.5°F), and oxygen saturation of 98% on room air. On physical examination, the patient is ill-appearing. His abdomen is severely tender to palpation and distended with no rebound or guarding. Pain is disproportionate to the exam findings. Rectal examination demonstrates bright red-colored stool. Abdominal X-ray is unremarkable. Stool culture was negative for *C. difficile*. A contrast-enhanced CT scan reveals segmental colitis involving the distal transverse colon. Which of the following is the most likely cause of this patient's symptoms?

Answer Choices:

A. Atherosclerosis B. Hypokalemia C. Bacterial infection D. Upper GI bleeding

Correct Answer: A. Atherosclerosis

Question 188

Clinical Scenario:

A 33-year-old man with HIV comes to the physician because of a nonproductive cough and shortness of breath for 3 weeks. He feels tired after walking up a flight of stairs and after long conversations on the phone. He appears chronically ill. His temperature is 38.5°C (101.3°F), and pulse is 110/min. Pulse oximetry on room air shows an oxygen saturation of 95%. Upon walking, his oxygen saturation decreases to 85%. Cardiopulmonary examination is normal. Laboratory studies show a CD4+ T-lymphocyte count of 176/mm³ (N > 500). Results of urine Legionella antigen testing are negative. A CT scan of the chest shows diffuse, bilateral ground-glass opacities. Microscopic examination of fluid obtained from bronchoalveolar lavage will most likely show which of the following findings?

Answer Choices:

A. Silver-staining, disc-shaped cysts B. Intracellular, acid-fast bacteria C. Septate, acute-branching hyphae D. Silver-staining, gram-negative bacilli

Correct Answer: A. Silver-staining, disc-shaped cysts

Question 189

Clinical Scenario:

A 24-year-old woman, G1P0, presents to her OB/GYN for her annual examination with complaints of painful cramps, abdominal pressure, and bloating with her cycle. She reports that she has not menstruated since her missed abortion requiring dilatation and curettage (D&C) seven months ago. She is sexually active with her husband and is not using any form of contraception. Her BMI is 29. At the clinic, her vitals are as follows: temperature, 98.9°F; pulse, 80/min; and blood pressure, 120/70 mm Hg. The physical examination is unremarkable. Thyroid-stimulating hormone, follicle-stimulating hormone, and prolactin concentrations are all within normal limits. The patient tests negative for qualitative serum beta-hCG. A progestin challenge test reveals no withdrawal bleeding. What is the most likely diagnosis?

Answer Choices:

A. Ectopic pregnancy B. Pelvic inflammatory disease C. Asherman syndrome D. Hypothalamic hypoenestrogenism

Correct Answer: C. Asherman syndrome

Question 190

Clinical Scenario:

Two days after being admitted for acute myocardial infarction, a 61-year-old man has sharp, substernal chest pain that worsens with inspiration and improves when leaning forward. Cardiac examination shows a scratchy sound best heard over the left sternal border. Histopathological examination of the affected tissue is most likely to show which of the following findings?

Answer Choices:

A. Neutrophilic infiltration B. Coagulative necrosis C. Collagenous scar tissue D. Granulation tissue with macrophages

Correct Answer: A. Neutrophilic infiltration

Question 191

Clinical Scenario:

A 24-year-old female comes to the physician for evaluation of a delayed menstrual period and intermittent lower abdominal pain for 2 days. Menarche occurred at the age of 12 years, and menses have occurred at regular 28-day intervals. Her last menstrual period was 7 weeks ago. Two years ago, she was treated for chlamydia infection. Pelvic examination shows a soft, mildly enlarged uterus. Endometrial biopsy shows decidualization of the endometrium without chorionic villi. Further evaluation of this patient is most likely to show which of the following findings?

Answer Choices:

A. Benign proliferation of myometrial smooth muscle B. Fertilized ovum outside the uterus C. Endometrial infiltration by plasma cells D. Empty ovum fertilized by two sperm

Correct Answer: B. Fertilized ovum outside the uterus

Question 192

Clinical Scenario:

Researchers are studying the relationship between heart disease and alcohol consumption. They review the electronic medical records of 500 patients at a local hospital during the study period and identify the presence or absence of acute coronary syndrome (ACS) and the number of alcoholic drinks consumed on the day of presentation. The researchers determine the prevalence of ACS and of alcoholic drink consumption. They correlate the relationship between these two variables and find that patients who reported no alcohol consumption or 1 drink only that day had a lower risk of acute coronary syndrome than patients who reported 2 or more drinks. Which of the following is the most accurate description of this study type?

Answer Choices:

A. Randomized controlled trial B. Case-control study C. Cross-sectional study D. Retrospective study

Correct Answer: C. Cross-sectional study

Question 193

Clinical Scenario:

A 17-year-old boy is brought to the physician by his father because of a 7-month history of fatigue, recurrent leg cramps, and increased urinary frequency. His pulse is 94/min and blood pressure is 118/85 mm Hg. Physical examination shows dry mucous membranes. Laboratory studies show: Serum Na⁺ 130 mEq/L K⁺ 2.8 mEq/L Cl⁻ 92 mEq/L Mg²⁺ 1.1 mEq/L Ca²⁺ 10.6 mg/dL Albumin 5.2 g/dL Urine Ca²⁺ 70 mg/24 h Cl⁻ 375 mEq/24h (N = 110–250) Arterial blood gas analysis on room air shows a pH of 7.55 and an HCO₃⁻ concentration of 45 mEq/L. Impaired function of which of the following structures is the most likely cause of this patient's condition?"

Answer Choices:

A. Ascending loop of Henle B. Distal convoluted tubule C. Collecting duct D. Proximal convoluted tubule

Correct Answer: B. Distal convoluted tubule

Question 194

Clinical Scenario:

A 16-year-old girl presents to the emergency department complaining of acute bilateral lower quadrant abdominal pain. She states she is nauseous and reports a 24-hour history of multiple episodes of vomiting. She admits to having unprotected sex with multiple partners. Her temperature is 102.0°F (38.9°C). Physical examination reveals bilateral lower quadrant tenderness. Bimanual pelvic exam reveals cervical exudate and cervical motion tenderness. Her β-HCG is within normal limits. Transvaginal ultrasound reveals a tubular complex lesion located in the right lower quadrant. Which of the following is the most appropriate initial step in the treatment of this patient?

Answer Choices:

A. Cefoxitin and doxycycline B. Levofloxacin and metronidazole C. Metronidazole D. Fluconazole

Correct Answer: A. Cefoxitin and doxycycline

Question 195

Clinical Scenario:

A 2-year-old boy is being evaluated for an autosomal-recessive condition that produces valine instead of glutamine in the β-globin gene. On further examination, the patient's X-ray showed a crew cut appearance of the skull. Which of the following statements about his condition is false?

Answer Choices:

A. Complications are due to vaso-occlusion B. Target cells are seen in blood smear C. An individual needs 2 defective β-globin genes to have the sickle cell trait D. Salmonella paratyphi can cause osteomyelitis in these patients

Correct Answer: C. An individual needs 2 defective β -globin genes to have the sickle cell trait

Question 196

Clinical Scenario:

A 16-year-old boy is brought to the physician by his parents because of a 6-month history of progressive fatigue and worsening shortness of breath on exertion. The parents report that the boy “has always been tired out a bit more easily than other kids.” The family recently immigrated to the United States from rural South Korea. Pulse oximetry on room air shows an oxygen saturation of 96% on bilateral index fingers. Radial pulses are bounding. There is mild bluish discoloration and bulbous enlargement of the distal toes bilaterally. Echocardiography is most likely to show which of the following?

Answer Choices:

A. Single overriding great vessel arising from the heart
B. Persistent blood flow between the pulmonary artery and descending aorta
C. Positioning of the ascending aorta directly over a ventricular septal defect
D. Abnormal narrowing of the aorta at the aortic isthmus

Correct Answer: B. Persistent blood flow between the pulmonary artery and descending aorta

Question 197

Clinical Scenario:

A 28-year-old man comes to the physician because of a 1-year history of chronic back pain. He explains that the pain started after getting a job at a logistics company. He does not recall any trauma and does not have morning stiffness or neurological symptoms. He has been seen by two other physicians for his back pain who did not establish a diagnosis. The patient also has abdominal bloating and a feeling of constipation that started 3 weeks ago. After doing extensive research on the internet, he is concerned that the symptoms might be caused by pancreatic cancer. He would like to undergo a CT scan of his abdomen for reassurance. He has a history of episodic chest pain, for which he underwent medical evaluation with another healthcare provider. Tests showed no pathological results. He does not smoke or drink alcohol. He reports that he is under significant pressure from his superiors due to frequent performance evaluations. He takes daily multivitamins and glucosamine to prevent arthritis. His vital signs are within normal limits. Examination shows a soft, non-tender, non-distended abdomen and mild bilateral paraspinal muscle tenderness. The remainder of the examination, including a neurologic examination, shows no abnormalities. Laboratory studies are within the reference range. An x-ray of the spine shows no abnormalities. Which of the following is the most likely explanation for this patient's symptoms?

Answer Choices:

A. Malignant neoplasm
B. Atypical depression
C. Somatic symptom disorder
D. Acute stress disorder

Correct Answer: C. Somatic symptom disorder

Question 198

Clinical Scenario:

A 26-year-old woman is brought to the emergency department after a motor vehicle accident. She was driving on the highway when she was struck by a van. At the hospital she was conscious but was bleeding heavily from an open wound in her left leg. Pulse is 120/min and blood pressure is 96/68 mm Hg. She receives 3 L of intravenous saline and her pulse slowed to 80/min and blood pressure elevated to 116/70 mm Hg. The next morning she is found to have a hemoglobin of 6.2 g/dL. Her team decides to transfuse 1 unit of packed RBCs. Twenty minutes into the transfusion she develops a diffuse urticarial rash, wheezing, fever, and hypotension. The transfusion is immediately stopped and intramuscular epinephrine is administered. Which of the following scenarios is most consistent with this patient's reaction to the blood transfusion?

Answer Choices:

A. A patient history of cardiovascular disease B. Facial twitching when the patient's cheek is tapped C. A patient history of frequent sinopulmonary infections D. Unsanitary blood product storage practices in the hospital

Correct Answer: C. A patient history of frequent sinopulmonary infections

Question 199

Clinical Scenario:

A 42-year-old woman comes to the physician because of a 2-month history of progressive muscular weakness. She has had difficulty climbing stairs, getting up from chairs, and brushing her hair. Her vital signs are within normal limits. Muscle strength is 2/5 with flexion of the hips and 3/5 with abduction of the shoulders. She is unable to stand up from her chair without the use of her arms for support. Laboratory studies show elevations in leukocyte count, erythrocyte sedimentation rate, and creatine kinase concentration. Histological evaluation of a biopsy specimen of the deltoid muscle is most likely to show which of the following?

Answer Choices:

A. Muscle fiber necrosis with rare inflammatory cells B. Sarcolemmal MHC-I overexpression with CD8+ lymphocytic infiltration C. Relative atrophy of type II muscle fibers with hypertrophy of type I muscle fiber D. Perimysial inflammation with perivascular CD4+ T lymphocytic infiltration

Correct Answer: B. Sarcolemmal MHC-I overexpression with CD8+ lymphocytic infiltration

Question 200

Clinical Scenario:

A 19-year-old man presents to an orthopedic surgeon to discuss repair of his torn anterior cruciate ligament. He suffered the injury during a college basketball game 1 week ago and has been using a knee immobilizer since the accident. His past medical history is significant for an emergency appendectomy when he was 12 years of

age. At that time, he said that he never wanted to have surgery again. At this visit, the physician explains the procedure to him in detail including potential risks and complications. The patient acknowledges and communicates his understanding of both the diagnosis as well as the surgery and decides to proceed with the surgery in 3 weeks. Afterward, he signs a form giving consent for the operation. Which of the following statements is true about this patient?

Answer Choices:

A. He cannot provide consent because he lacks capacity B. He has the right to revoke his consent at any time
C. His consent is invalid because his decision is not stable over time D. His parents also need to give consent to this operation

Correct Answer: B. He has the right to revoke his consent at any time

Question 201

Clinical Scenario:

A 21-year-old male presents to your office with hematuria 3 days after the onset of a productive cough and fever. Following renal biopsy, immunofluorescence shows granular IgA deposits in the glomerular mesangium. Which of the following do you suspect in this patient?

Answer Choices:

A. Lipoid nephrosis B. Berger's disease C. Poststreptococcal glomerulonephritis D. HIV infection

Correct Answer: B. Berger's disease

Question 202

Clinical Scenario:

A 23-year-old man comes to the emergency department because of a rash on his neck and back for the past 6 hours. He says that he first noticed some reddening of the skin on his back the previous evening, which turned into a blistering, red rash overnight. He went surfing the previous day and spent 5 hours at the beach. He reports having applied at least 1 oz of water-resistant SPF 30 sunscreen 30 minutes before leaving his home. His vitals are within normal limits. Physical examination shows erythema of the skin over the upper back and dorsum of the neck, with 3 vesicles filled with clear fluid. The affected area is edematous and tender to touch. Which of the following recommendations is most appropriate to prevent a recurrence of this patient's symptoms in the future?

Answer Choices:

A. Reapply sunscreen after water exposure B. Use SPF 50 sunscreen C. Apply at least 3 oz of sunscreen D. Use waterproof sunscreen

Correct Answer: A. Reapply sunscreen after water exposure

Question 203

Clinical Scenario:

A 2-month-old boy presents to the clinic with his mother for evaluation of crusty, greasy patches on the skin of the scalp that appeared 1 week ago. The mother states that the patient has been acting normally and is feeding well. She had a vaginal birth with no complications. On examination, the patient is smiling and playful in his mother's arms. He can hold his head up and focus on faces and is happily gurgling. Vital signs are stable and weight, length, and head circumference measurements are all within normal limits. The skin on the scalp appears greasy, with yellow, scaly patches and evidence of inflammation. What is the most likely diagnosis?

Answer Choices:

A. Seborrheic dermatitis B. Atopic dermatitis C. Impetigo D. Chickenpox

Correct Answer: A. Seborrheic dermatitis

Question 204

Clinical Scenario:

A 15-year-old boy is brought to the physician because his urine has been pink since that morning. During the past 2 days, he has had a sore throat and difficulty swallowing. He also reports having a low-grade fever. He has no pain with urination or changes in urinary frequency. He has had 2 similar episodes involving a sore throat and pink urine over the past 2 years. His older sister has systemic lupus erythematosus. His temperature is 38.3°C (101°F), pulse is 76/min, and blood pressure is 120/80 mm Hg. Oral examination shows an erythematous pharynx and enlarged tonsils. The remainder of the examination shows no abnormalities. Laboratory studies show: Leukocyte count 20,000/mm³ Serum Urea nitrogen 8 mg/dL Creatinine 1.4 mg/dL Urine Blood 3+ Protein 1+ RBC 15–17/hpf with dysmorphic features RBC casts numerous Ultrasound of both kidneys shows no abnormalities. A renal biopsy is most likely to show which of the following findings?"

Answer Choices:

A. Splitting of the glomerular basement membrane B. Crescents of fibrin and plasma proteins C. IgA mesangial deposition D. Granular deposits of IgG, IgM, and C3 complement "

Correct Answer: C. IgA mesangial deposition

Question 205

Clinical Scenario:

A 6-year-old girl is brought to the physician because of a generalized pruritic rash for 3 days. Her mother has noticed fluid oozing from some of the lesions. She was born at term and has been healthy except for an episode of bronchitis 4 months ago that was treated with azithromycin. There is no family history of serious illness. Her immunization records are unavailable. She attends elementary school but has missed the last 5 days. She appears healthy. Her temperature is 38°C (100.4°F). Examination shows a maculopapular rash with crusted

lesions and vesicles over the entire integument, including the scalp. Her hemoglobin concentration is 13.1 g/dL, leukocyte count is 9800/mm³, and platelet count is 319,000/mm³. Which of the following is the most appropriate next best step?

Answer Choices:

A. Vitamin A therapy B. Rapid strep test C. Tzanck test D. Calamine lotion

Correct Answer: D. Calamine lotion

Question 206

Clinical Scenario:

A 51-year-old woman comes to the physician because of a 1-year history of occasional discoloration and tingling in her fingers. She has no history of major medical illness and takes no medications. Examination of the hands and fingers shows thickened, waxy skin and several firm white nodules on the fingertips. Further evaluation of this patient is most likely to show which of the following findings?

Answer Choices:

A. Upper eyelid rash B. Telangiectasia C. Endocardial immune complex deposition D. Interstitial lung disease

Correct Answer: B. Telangiectasia

Review Summary

Category: Clinical Findings **Questions Reviewed:** ____ / 206

Overall Assessment: - Correctly categorized: _ questions - Miscategorized: questions - Ambiguous: __ questions

General Comments:

Reviewer Name: _____

Date: _____

Credentials: _____